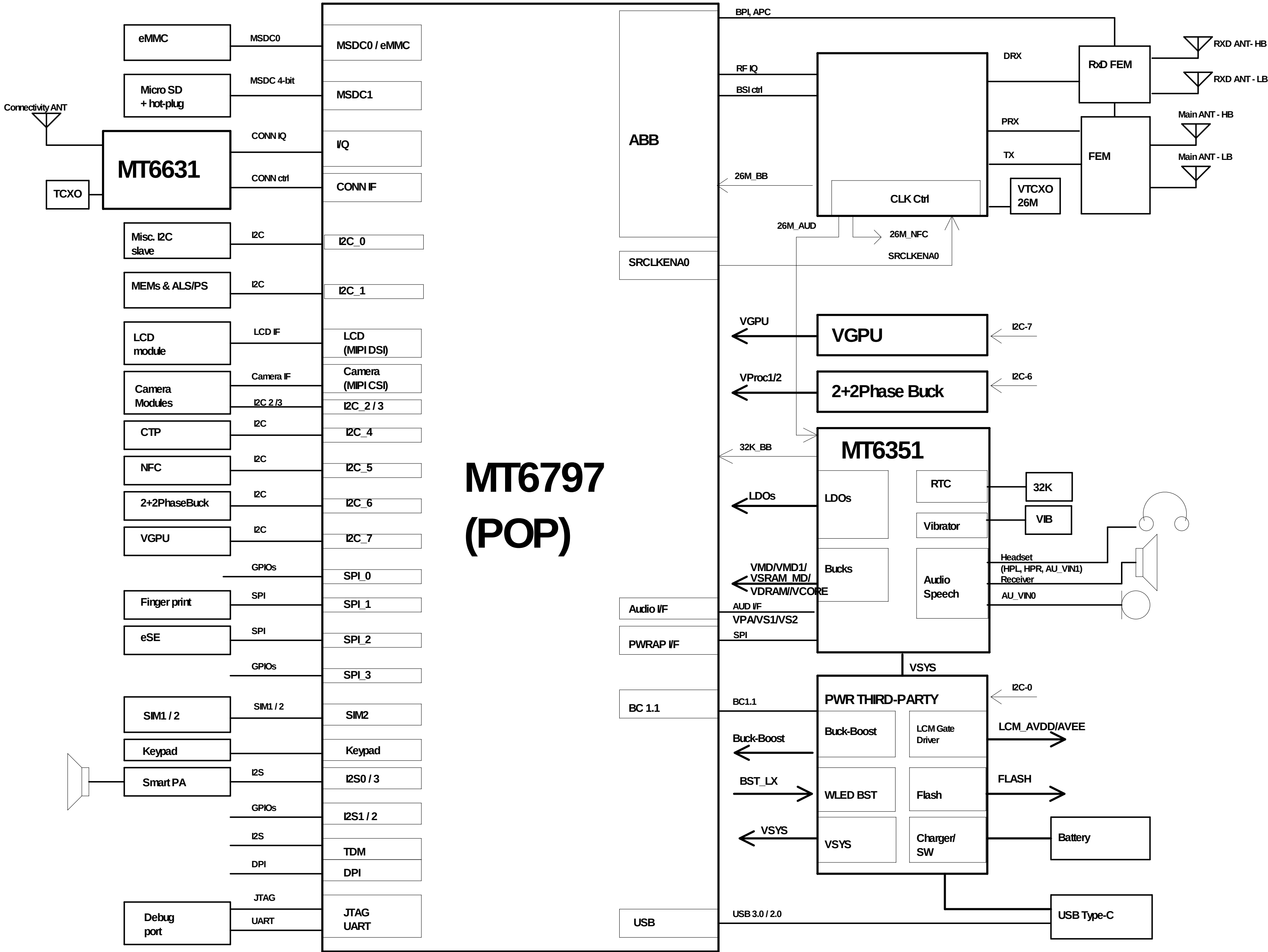


BLOCK_DIAGRAM

Project : MT6797

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 00_BLOCK_DIAGRAM		VERSION: V1.1	SHEET: 1 OF 23
CHECKED	< >	DATED	< >				

I2C_ID_OVERVIEW

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

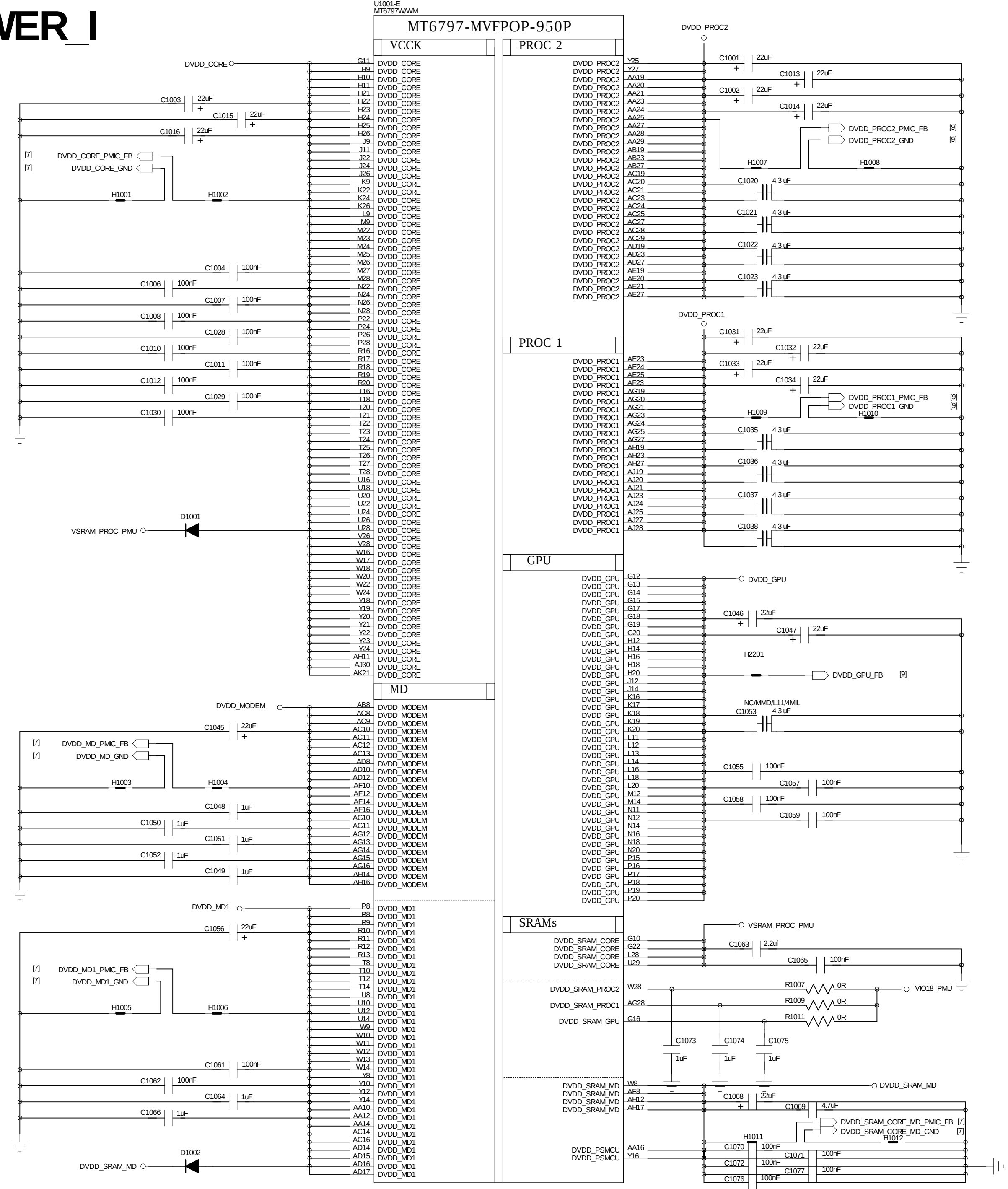
I2C	Function	I2C Spec.[1]	Budgeted Timing	I2C Slave Address (7-bit mode)
I2C-0 *[2]	SW charger	400 Kbps		bq25892/ SW charger I2C address: 0X6B (Write:0xD6, Read:0xD7)
	LCM Gate Driver	400 Kbps		NT50358 / LCM Gate Driver I2C address: 0X3E (Write:0x7C, Read:0x7D)
	Buck-boost	400 Kbps		FAN49101 / Buck-boost I2C address: 0X70 (Write:0xE0, Read:0xE1)
	Flash LED Driver	400 Kbps		KTD2686 / Flash LED I2C address: 0X63 (Write:0xC6, Read:0xC7)
	Speaker Amp.	400 Kbps		TFA9891UK / Speaker AMP I2C Address: 0x34 (Write:0x68, Read:0x69) TFA9891UK / Speaker AMP I2C Generic Address: 0x0E (Write:0x1C, Read:0x1D)
I2C-1 *[2]	ALS Sensor	400 Kbps		VL6180X:I2C ADDRESS:0x52(Write)/0x53(Read)
	AL/PS Sensor	400 Kbps		LTR-PA22A00001:I2C address: Write:3Ch Read:3Dh
	M-Sensor	400 Kbps		MMC35240PJ:I2C ADDRESS:0x60(Write)/0x61(Read) AF8133I/ I2C Address : Write:0x3C, Read : 0x3D/Write:0x38 , Read:0x39
	Gyro Sensor	400 Kbps		BM160 / I2C Address : Write:0xD0, Read : 0xD1/Write:0xD2 , Read:0xD3
I2C-2	Rear camera	400 Kbps	Yes.	IMX230/ Rear camera I2C address: 0X10 (Write:0x20, Read:0x21) if SLESEL = Low or NC. IMX230/ Rear camera I2C address: 0X1A (Write:0x34, Read:0x35) if SLESEL =High Default
	Rear camera's AF	400 Kbps		LC898132AXD / AF driver I2C address: (Write:0x7C Read:0x7D)
	Rear camera's EEPROM	400 Kbps		GT24C64A / AF driver I2C address: 0X50 (Write:0xA0 Read:0xA1)
I2C-3	Front camera	400 Kbps	Yes.	Front camera (S5K3L8YX) I2C address: (Write:0x5A, Read:0x5B)
I2C-4	CTP	400 Kbps	Yes.	GT915L / CTP I2C address: 0X5D (Write:0xBA, Read:0xBB) or 0X14 (Write:0x28, Read:0x29)
I2C-5	VDRAM Buck	400 Kbps	Yes.	FAN53526 / Buck I2C address: 0X60 (Write:0xC0, Read:0xC1)
I2C-6	VPROC buck	3.4 Mbps	Yes.	DA9214 / 2+2Phase Buck I2C address: Register Page 0 address = 0X68 (Write:0xD0, Read:0xD1); Register Page 2 address = 0X69 (Write:0xD2, Read:0xD2)
I2C-7	VGPU Buck	3.4 Mbps	Yes.	FAN53200 / Buck I2C address: 0X60 (Write:0xC0, Read:0xC1)

Note 1: I2C Spec. : Standard mode (100 kbps) and Fast mode (400 kbps), Fast mode Plus (1 Mbps) and High-speed mode (3.4 Mbps)

Note 2: For MEMs sensor/sensor hub application, these I2C slave devices must be connected to I2C0 or I2C-1.

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 01_I2C_ID_OVERVIEW		VERSION: V1.1	SHEET: 2 OF 23
CHECKED	< >	DATED	< >				

BB_POWER_I



SN2

1 SN

SH1001

SH+BB-H972

SH1002

SH+RF-H972

SH1003

SH+PA-H972

SH1004

SH+MEM-H972

SH1005

SH+MT6831-H972

SH1006

SH+CHG-H972

SH1007

SH+LCM-H972

SH1008

SH+CAM-H972

HP1001

BOOST1001

HP1002

CAMERA_BOOST2

HP1003

MARK1001

HP1004

MARK1002

HP1005

SN1001

HP1006

WPL1001

HP1007

WPL1002

HP1008

HOLE-PTH-V2.6D3.5

HP1009

TRANSSION-LOGO

HP1010

TRANSSION-3.5H

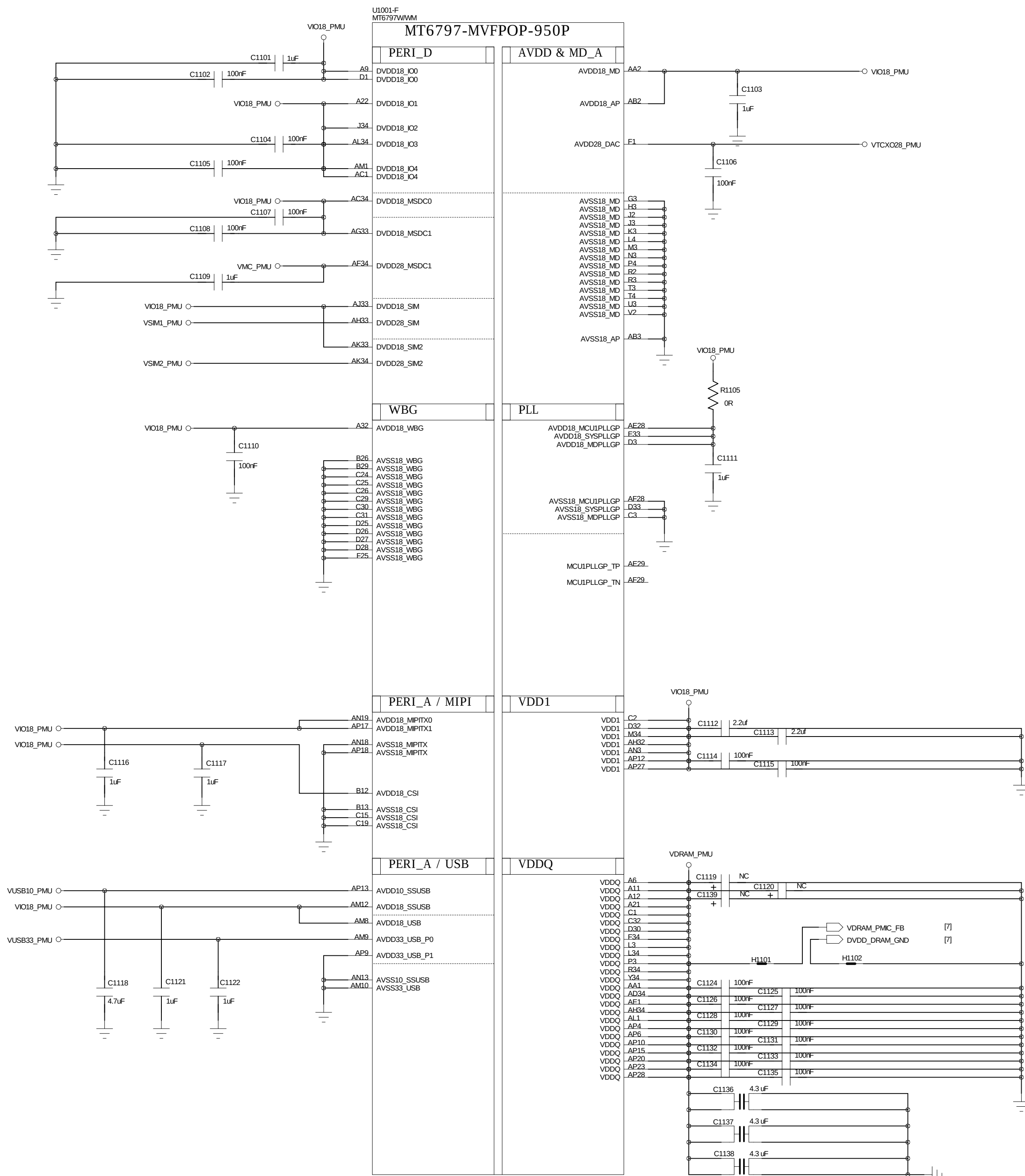
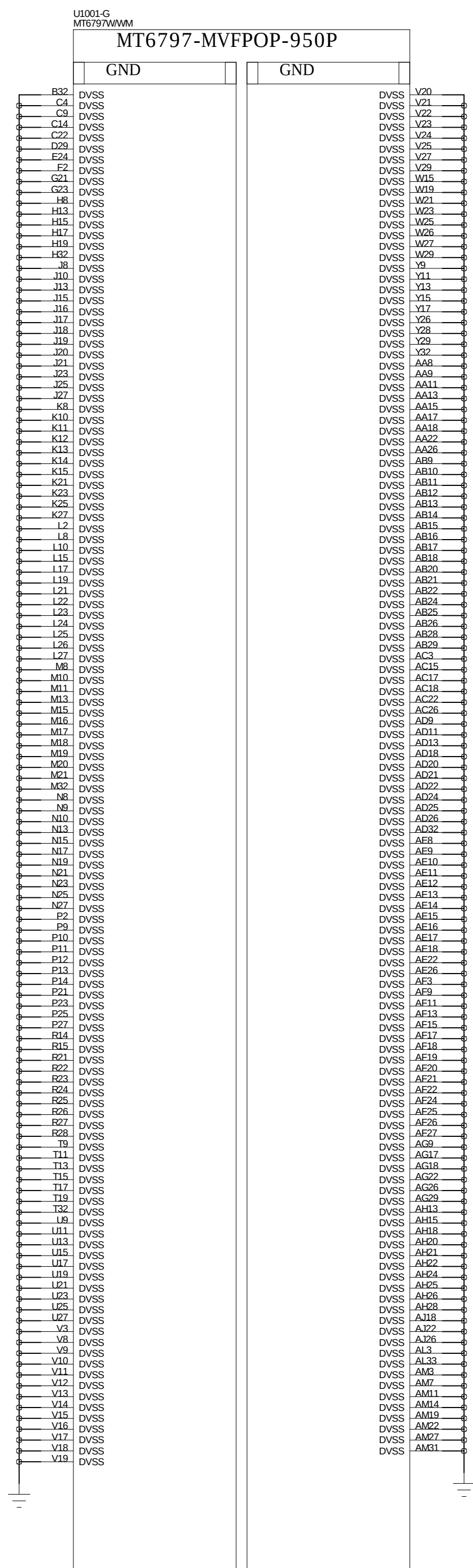
HP1007

TRANSSION-3.5H

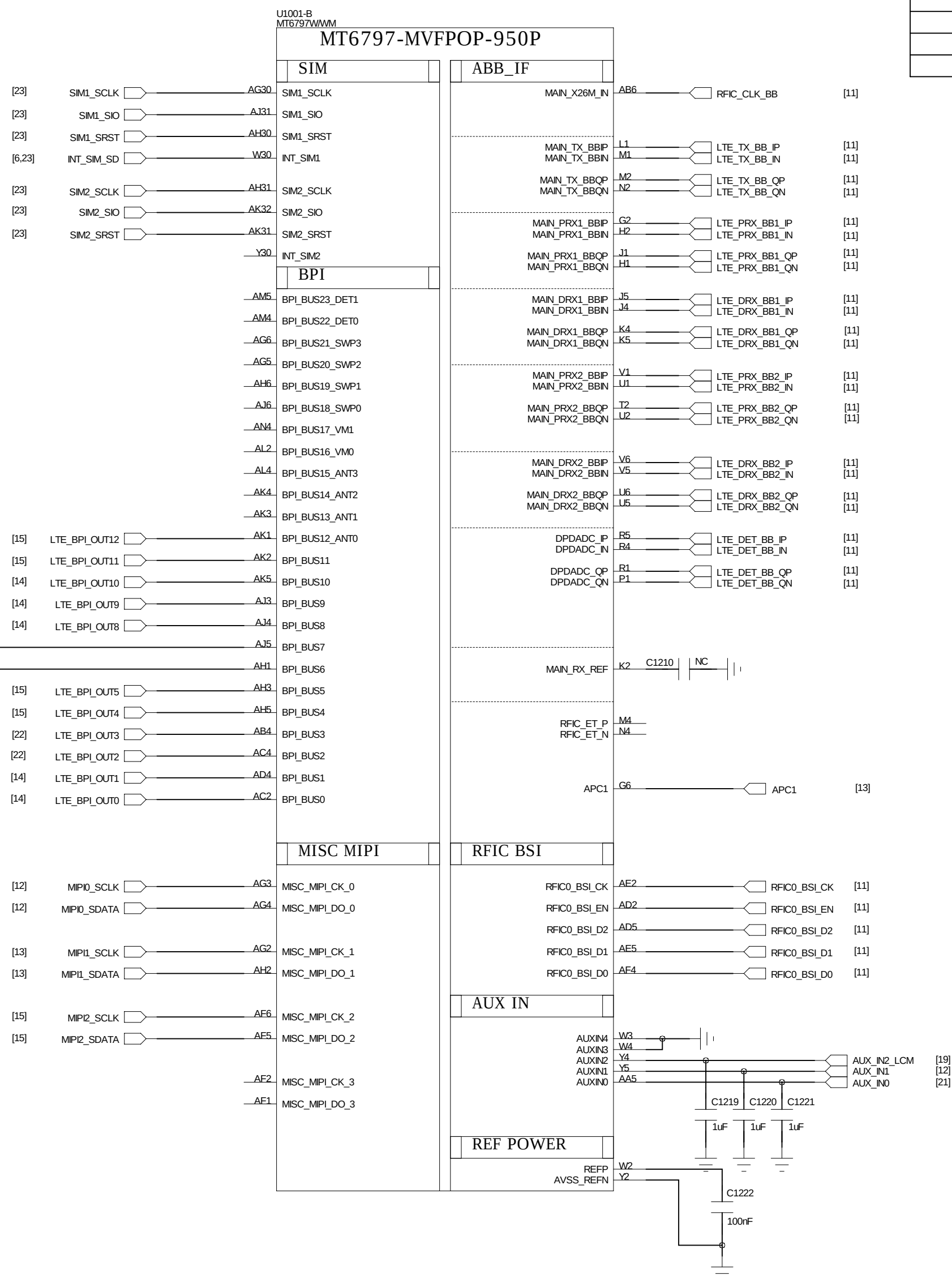
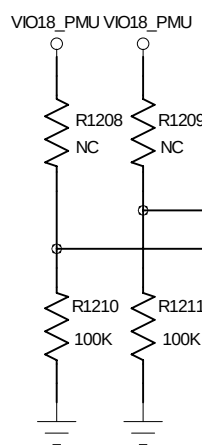
COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 10_BB_POWER_I		VERSION: V1.1	SHEET: 3 OF 23
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BB_POWER II

REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:

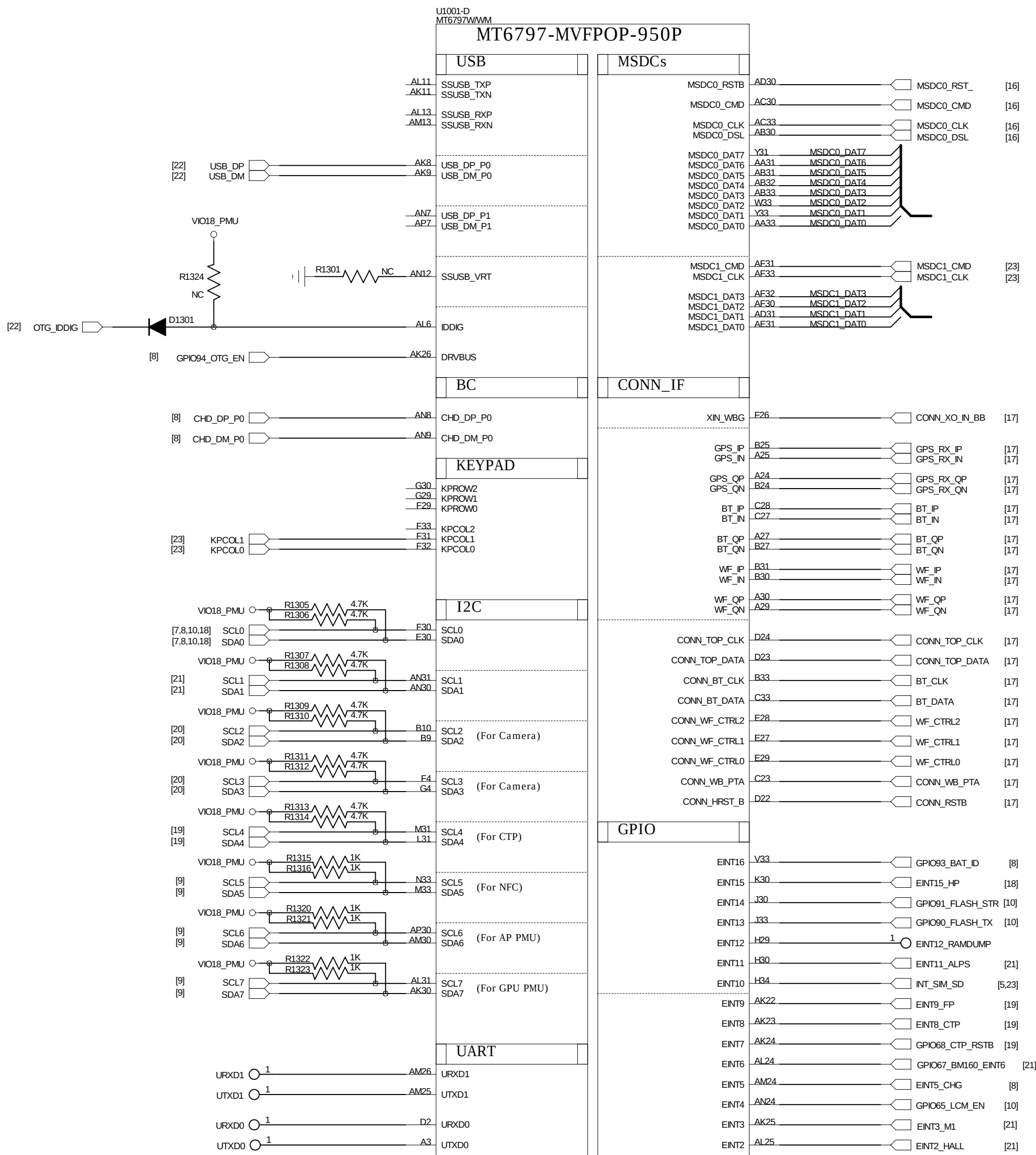
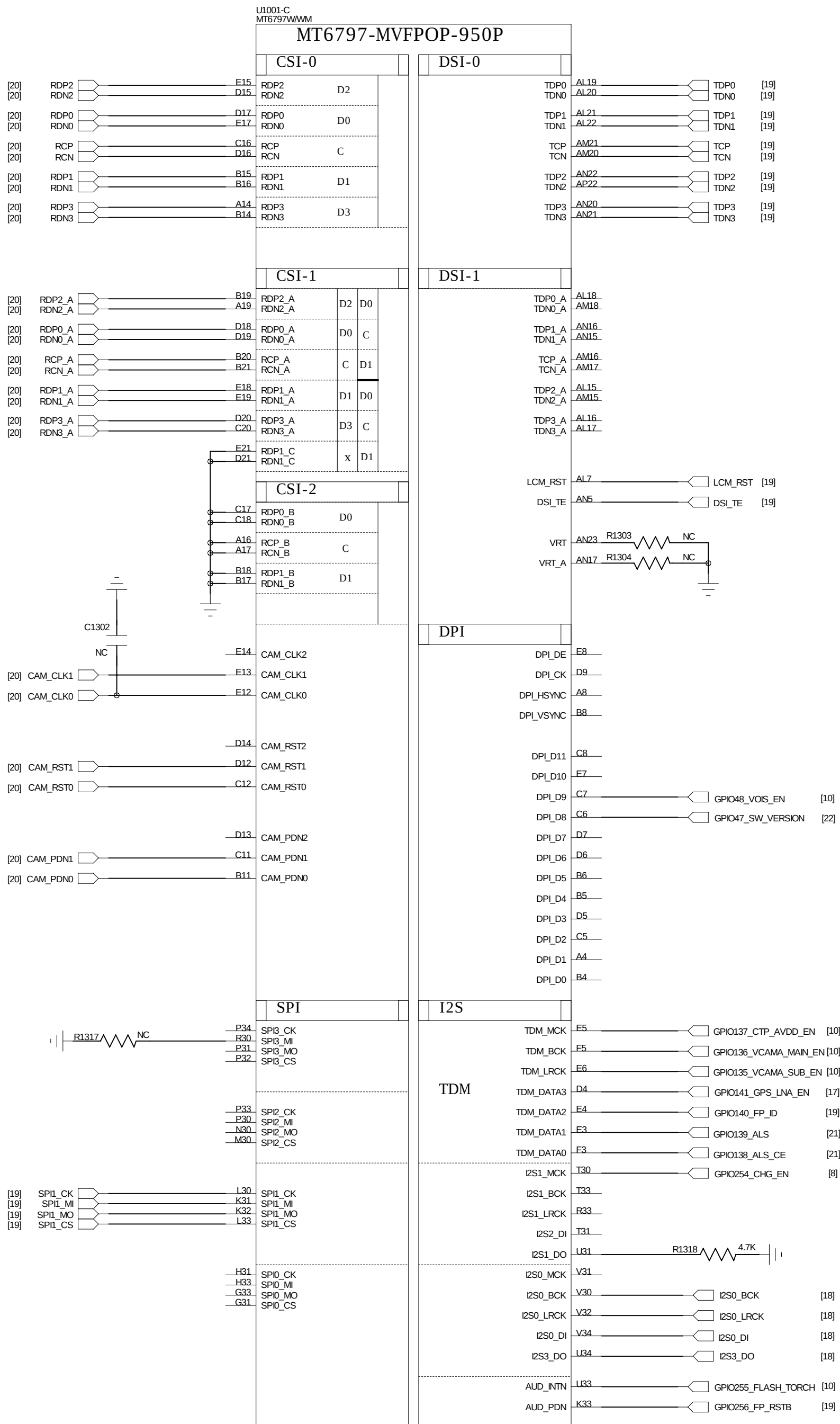


COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972	Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 11_BB_POWER_2	VERSION: V1.1	SHEET: 4 OF 23
CHECKED	< >	DATED	< >			



COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972	Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 12_BB_I	VERSION: V1.1	SHEET: 5 OF 23
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LTR	ECO NO.	APPROVED:	DATE:

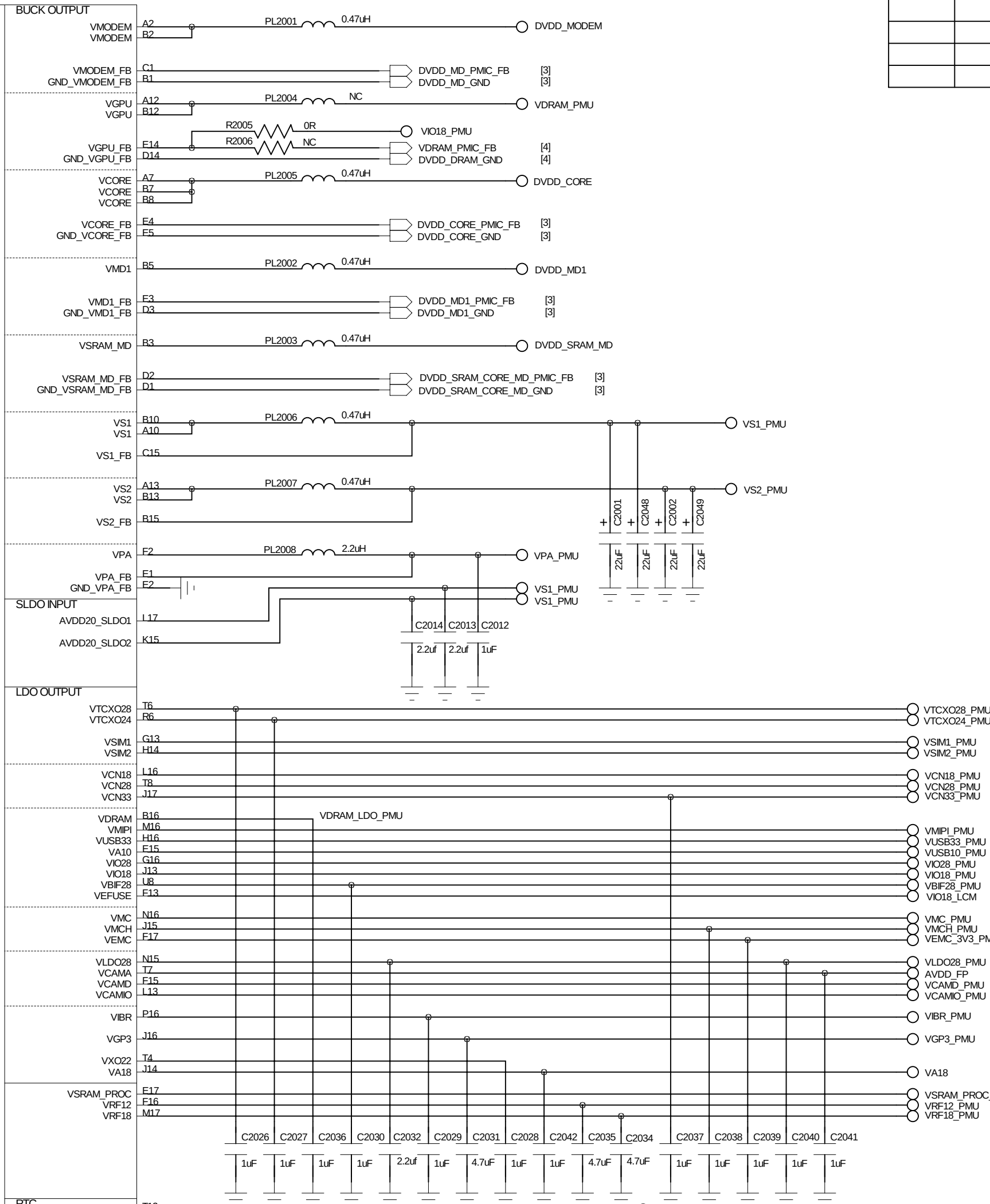
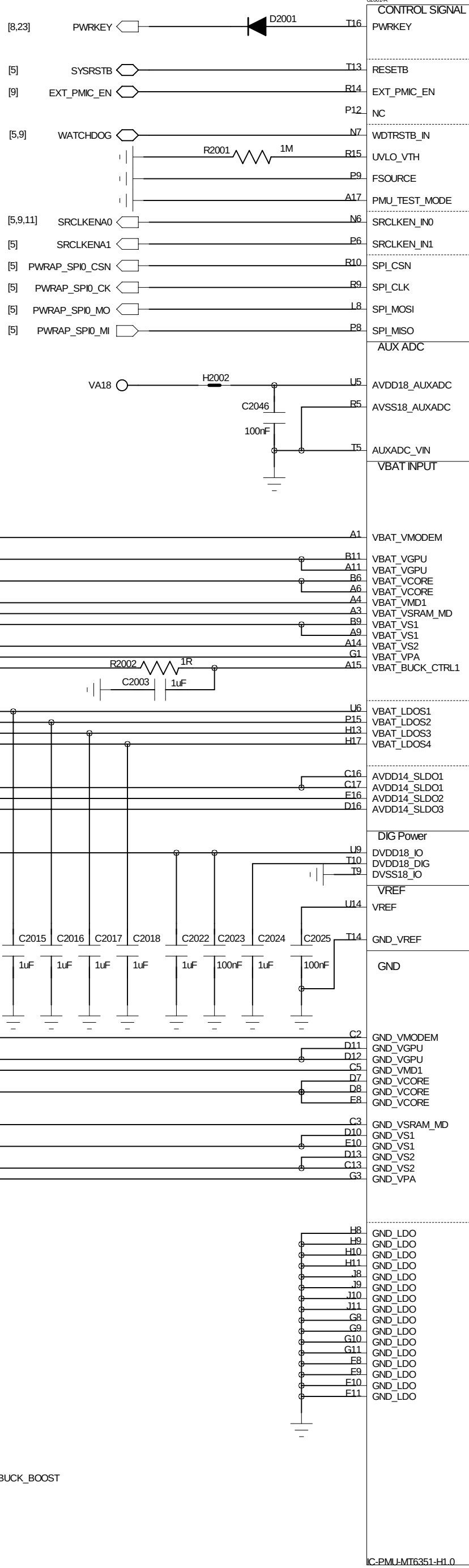


COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972	Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 13_BB_II	VERSION: V1.1	SHEET: 6 OF 23
CHECKED	< >	DATED	< >			

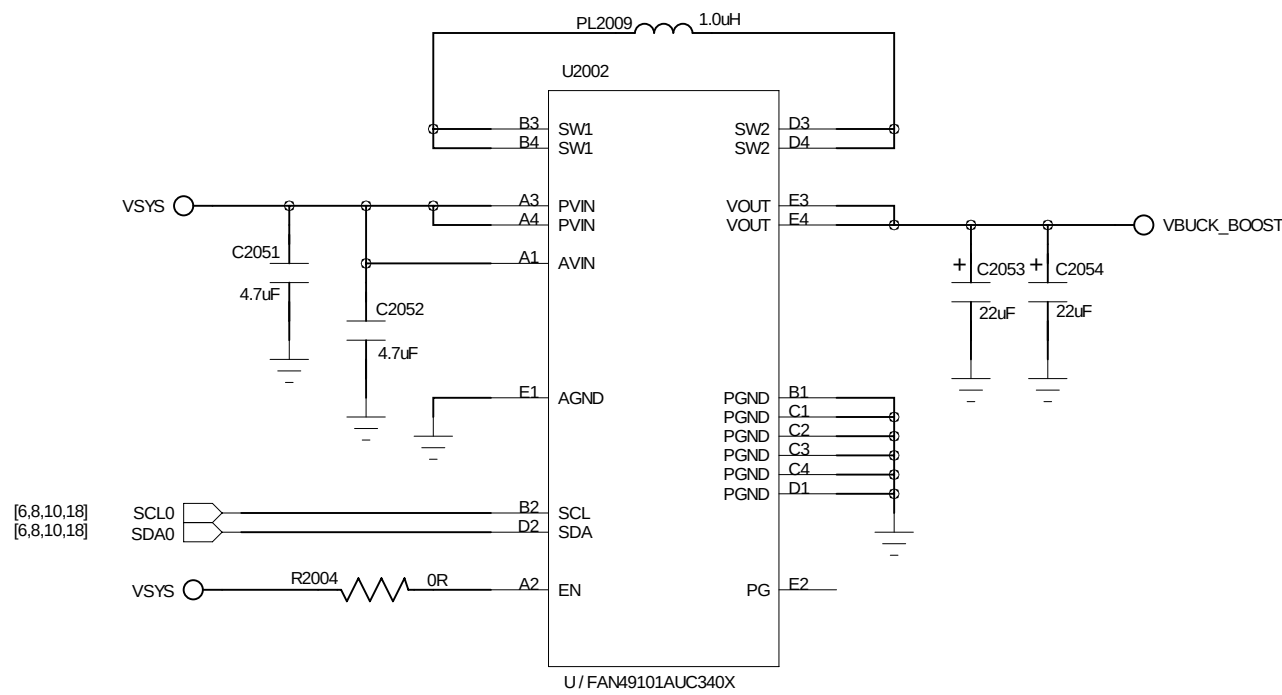
POWER_MT6351_I

PMU_PART_I

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



Companion Buck/Boost



COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 20_POWER_MT6351_I		VERSION: V1.1	SHEET: 7 OF 23
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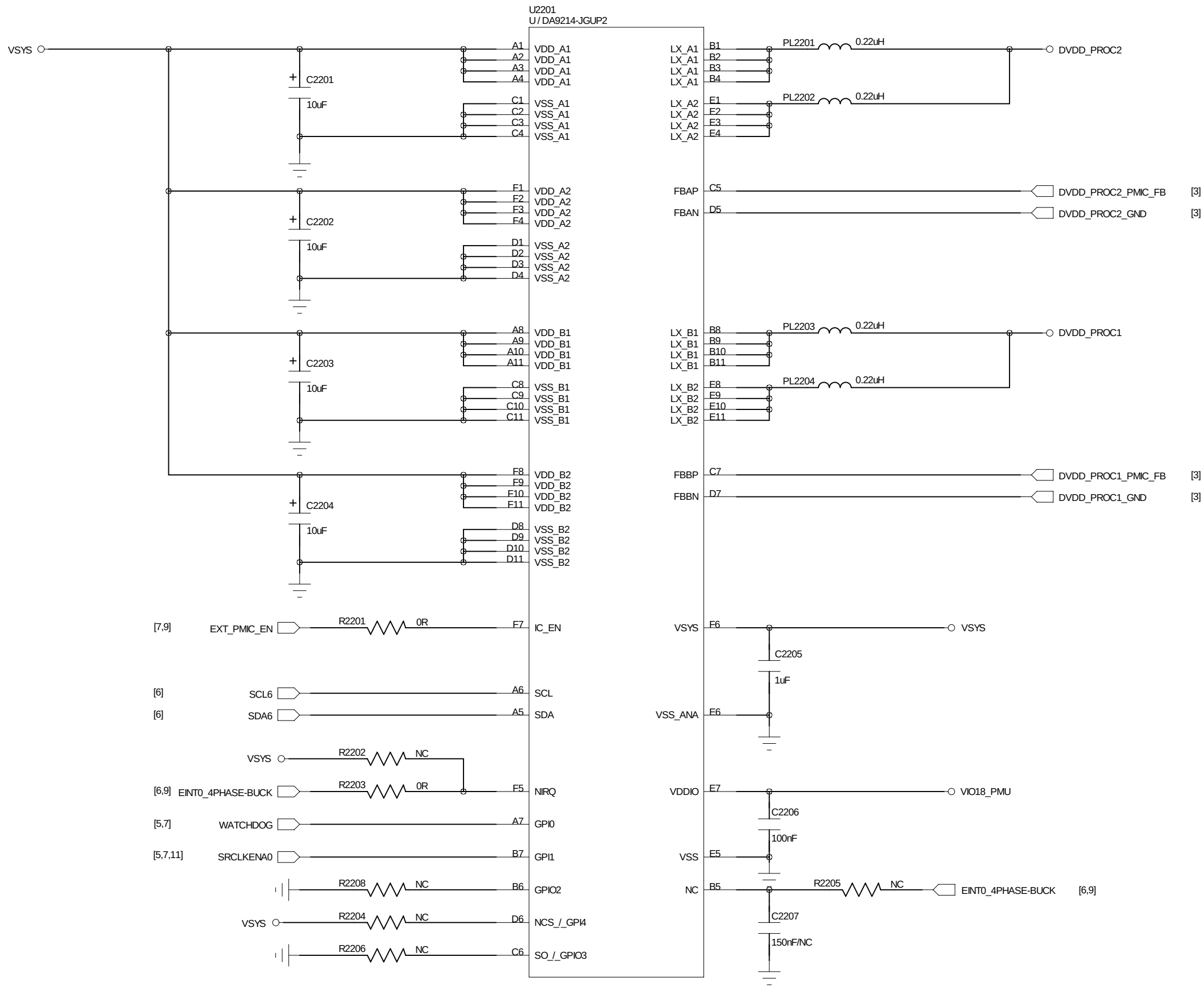
REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972	Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 21_POWER_MT6351_II_CHG	VERSION: V1.1	SHEET: 8 OF 23
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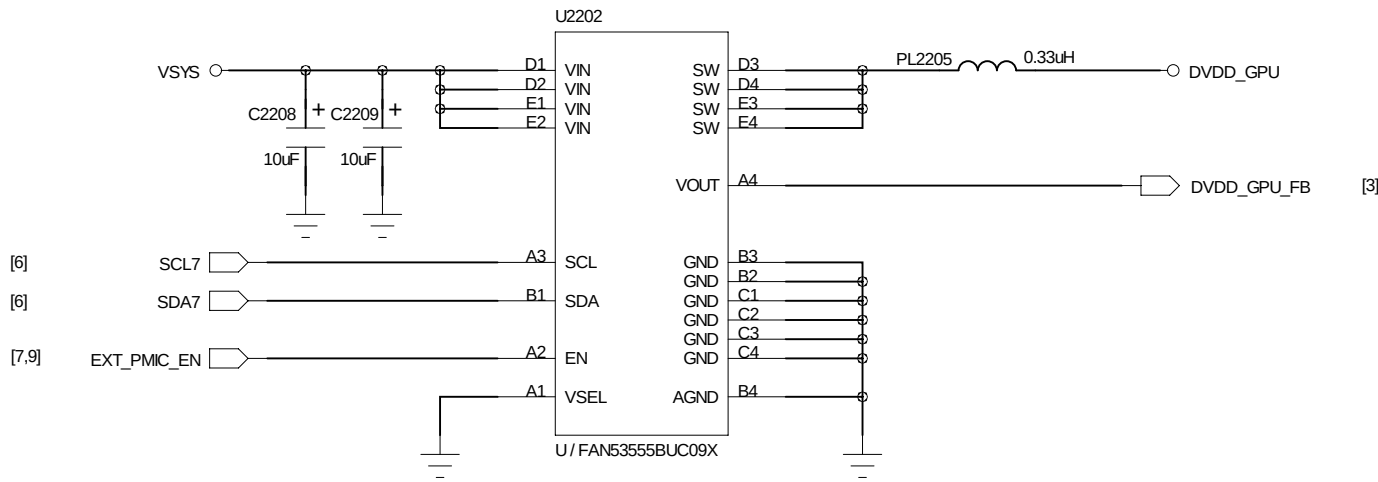
POWER_VPROCS_VGPU_VM

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

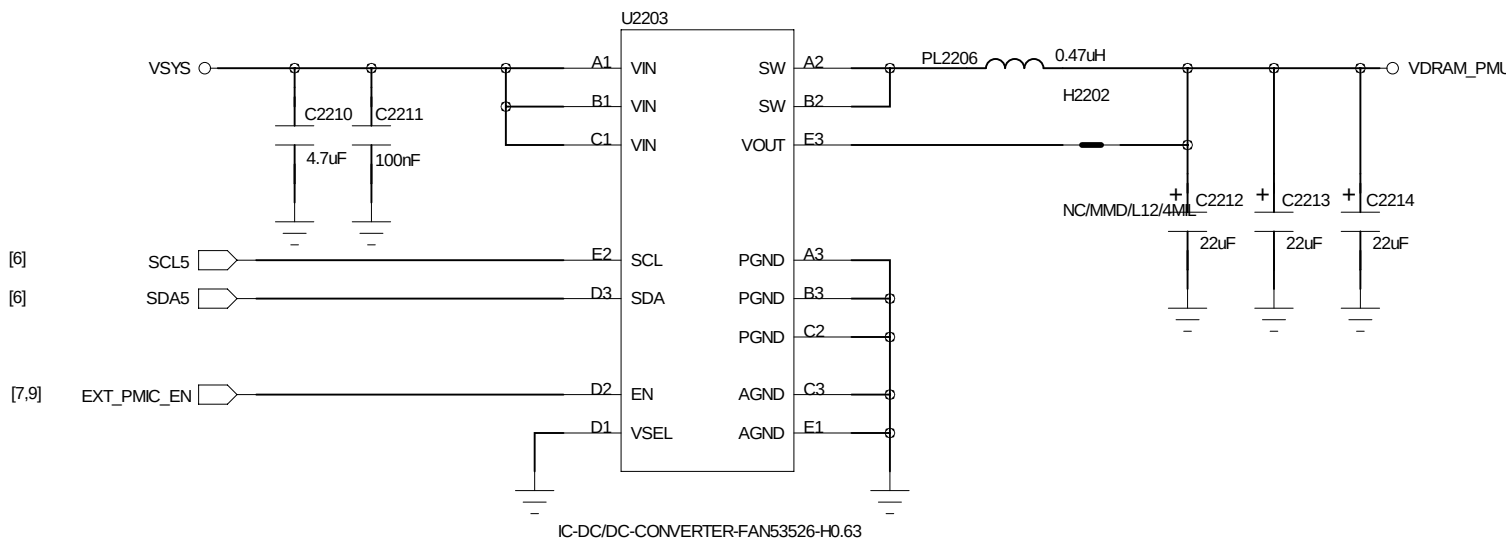
4-Phase Buck



1-Phase Buck



Buck for VDRAM

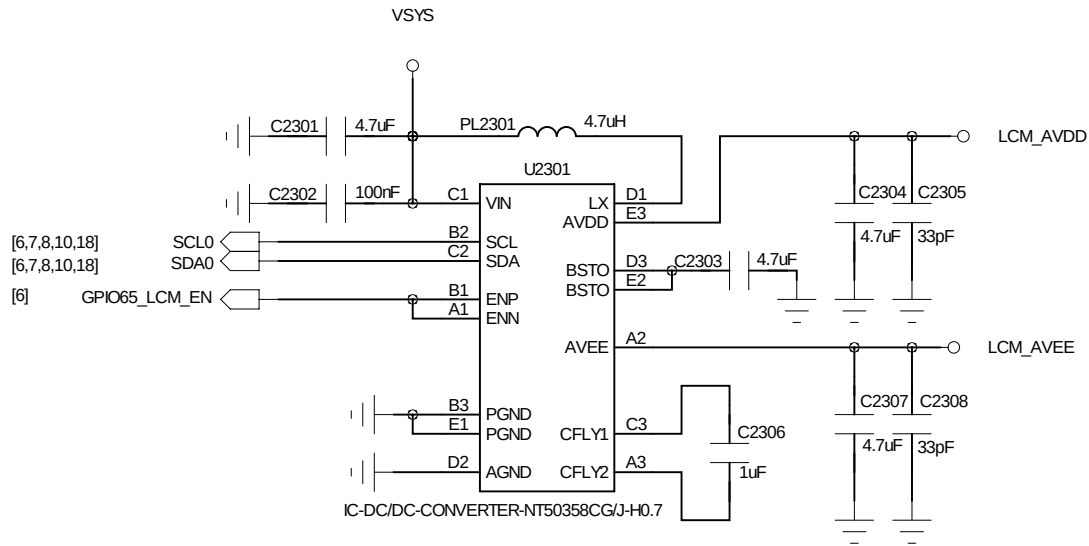


COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 22_POWER_VPROCS_VGPU_VM		VERSION: V1.1	SHEET: 9 OF 23
CHECKED	< >	DATED	< >				

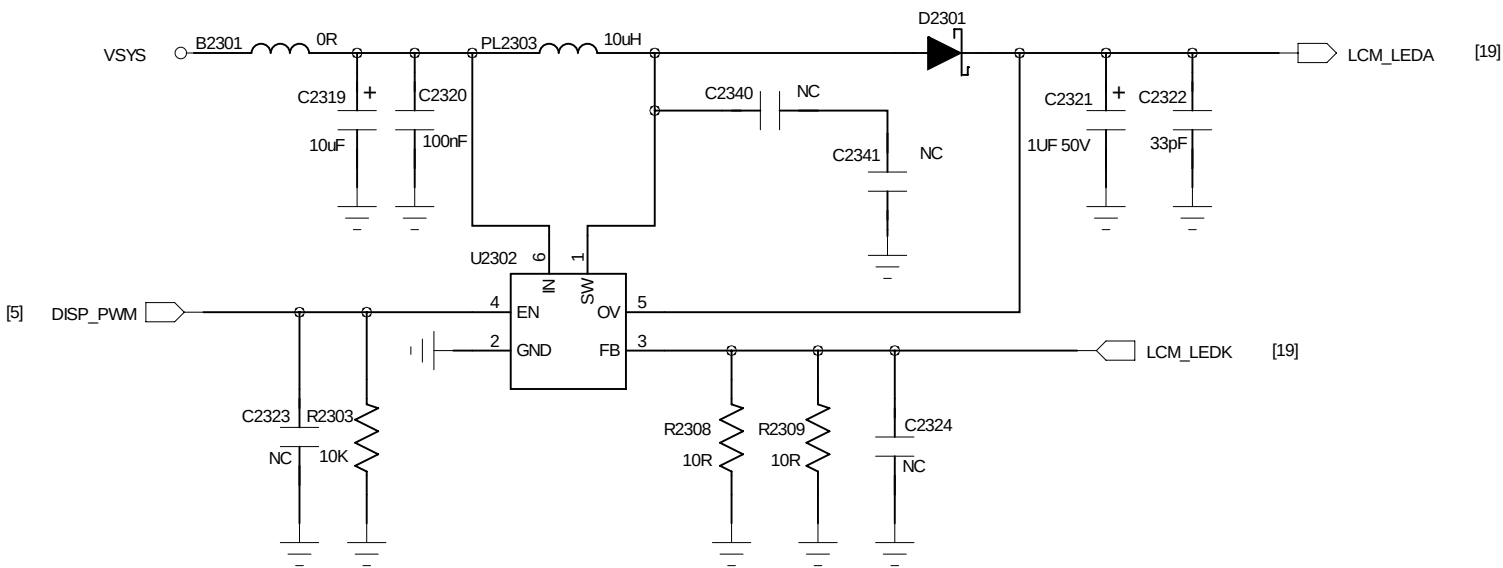
POWER_THIRD-PART

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

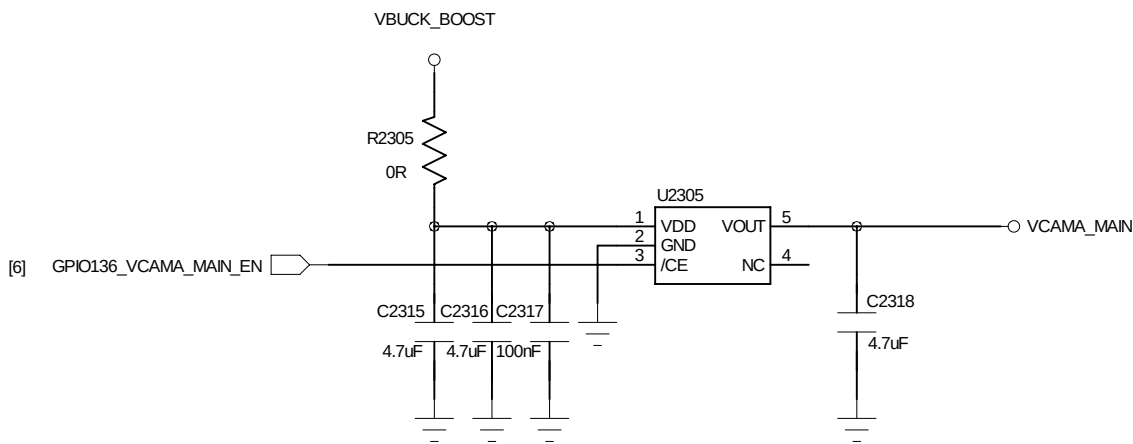
LCM BOOST



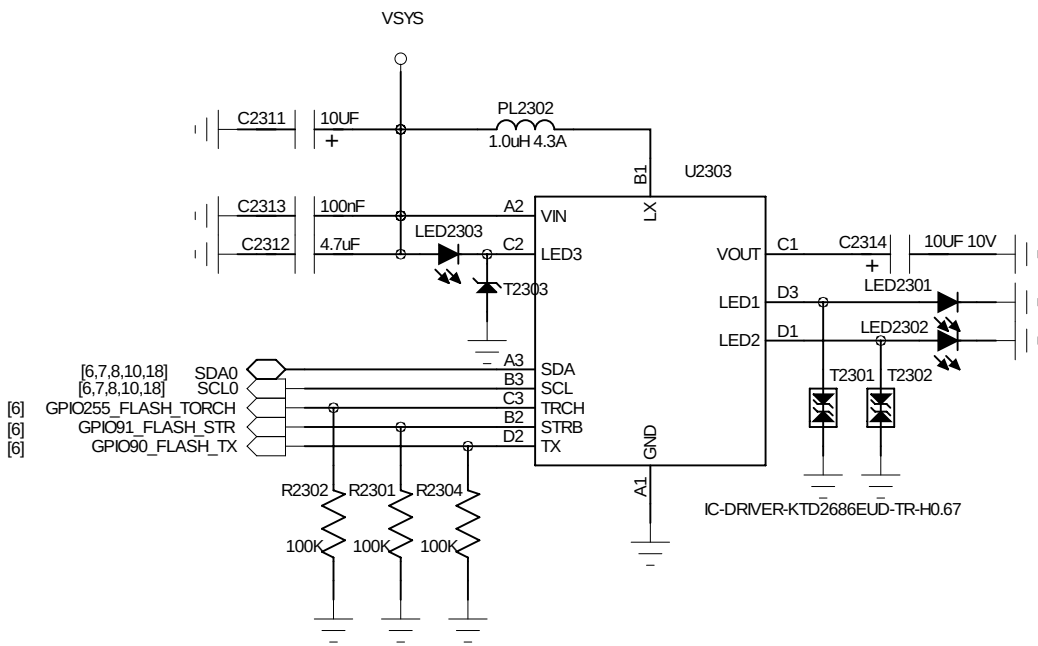
LCM Backlight LED Driver



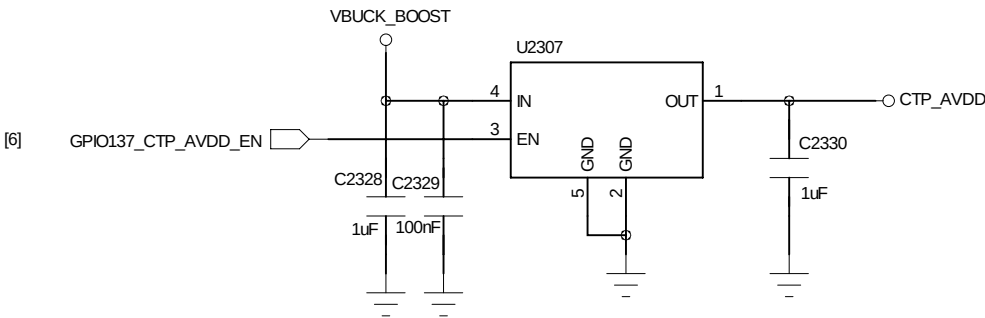
2.5V_VCAMA for MAIN CAM



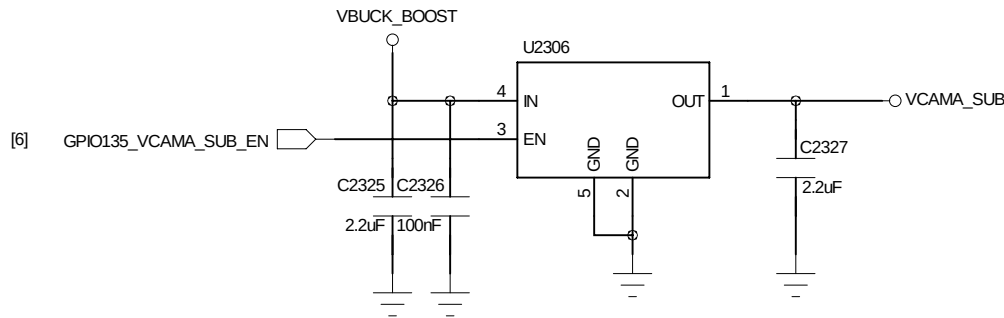
CAMERA FLASH LED



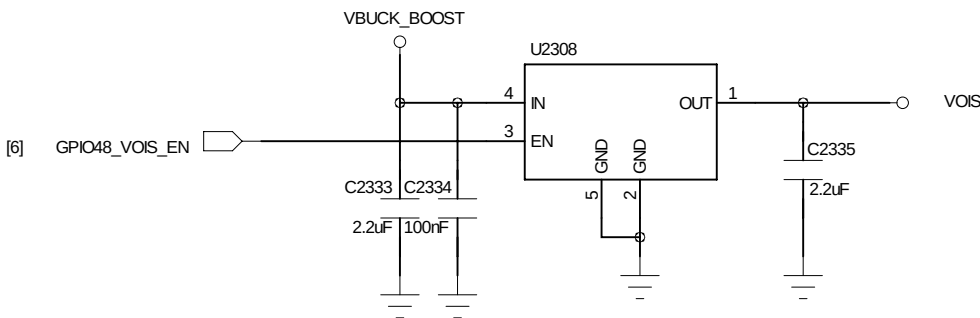
3.0V_AVDD for CTP



2.8V_VCAMA for SUB CAM



2.8V_OIS for MAIN CAM



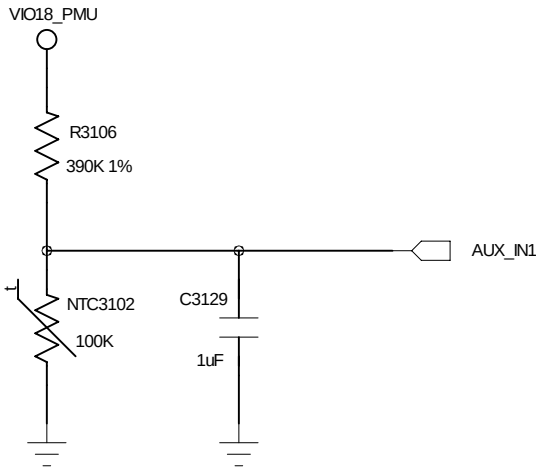
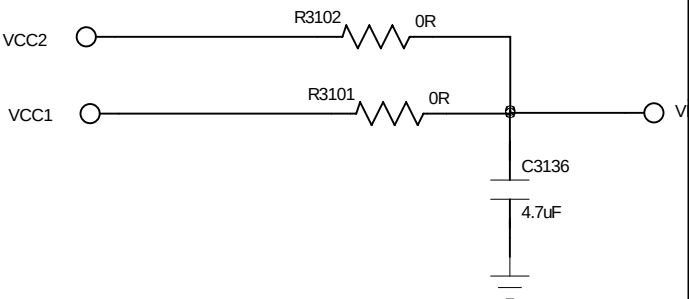
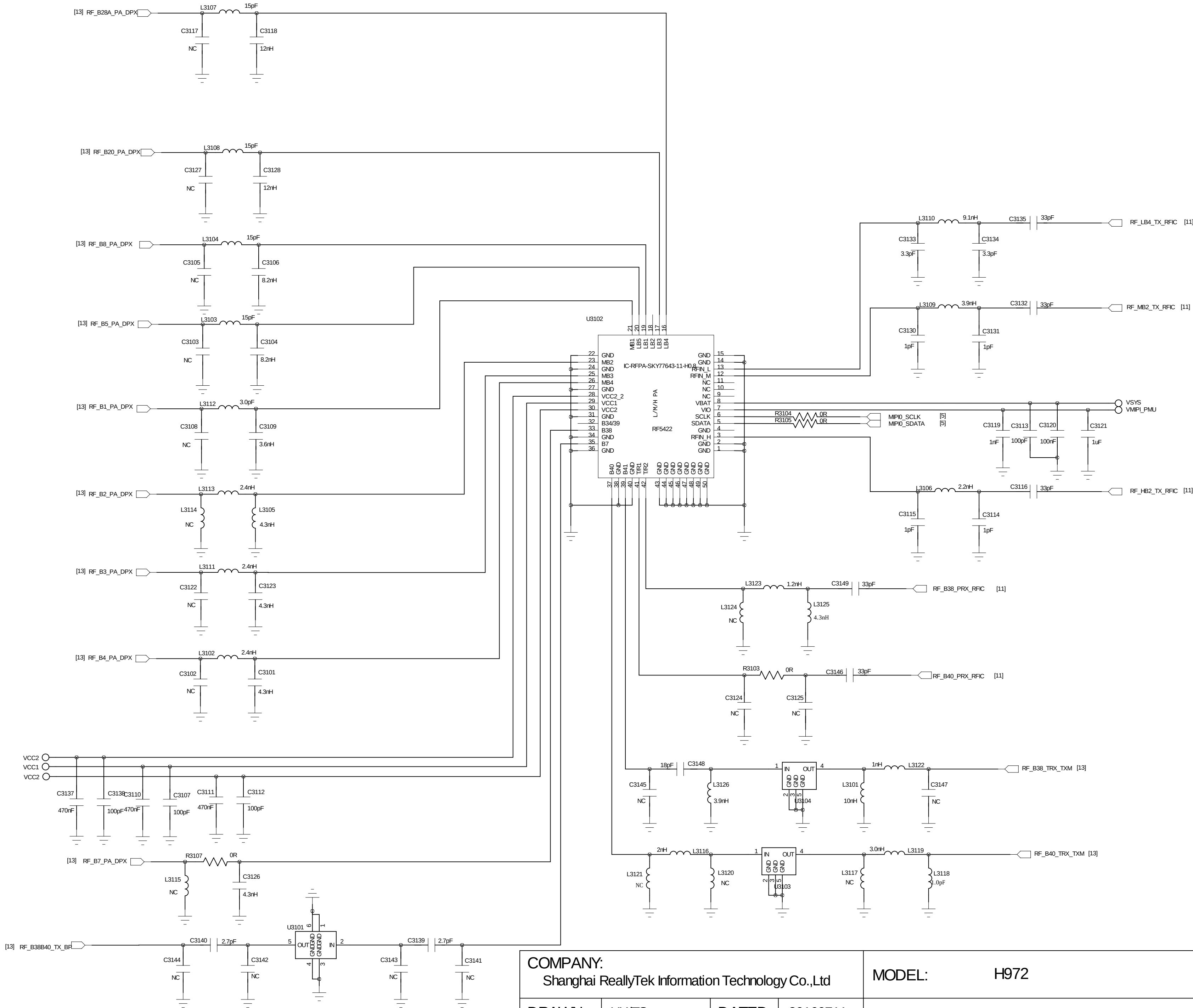
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DRAWN	HX/TS	DATED	20160711	TITLE: 23_POWER_THIRD-PARTY		VERSION: V1.1	SHEET: 10OF 23
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REVISION RECORD			
LTR	ECO NO:	APPROVED:	DATE:



RF_MT6176_RF_TX

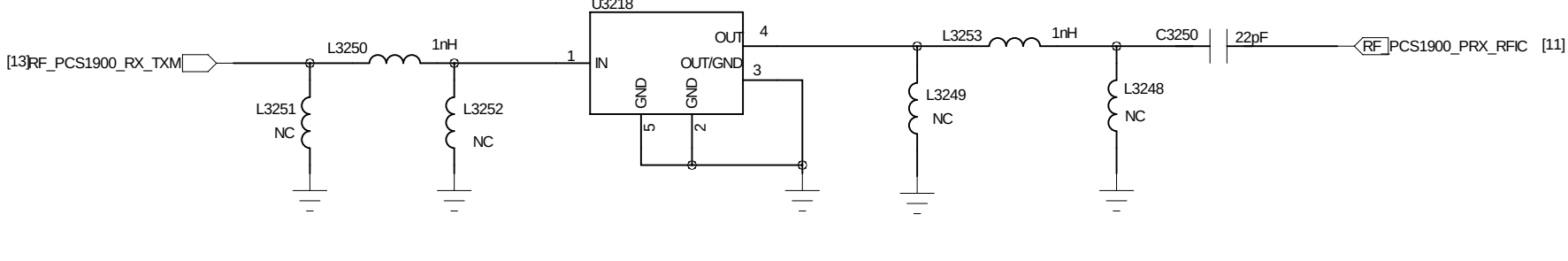
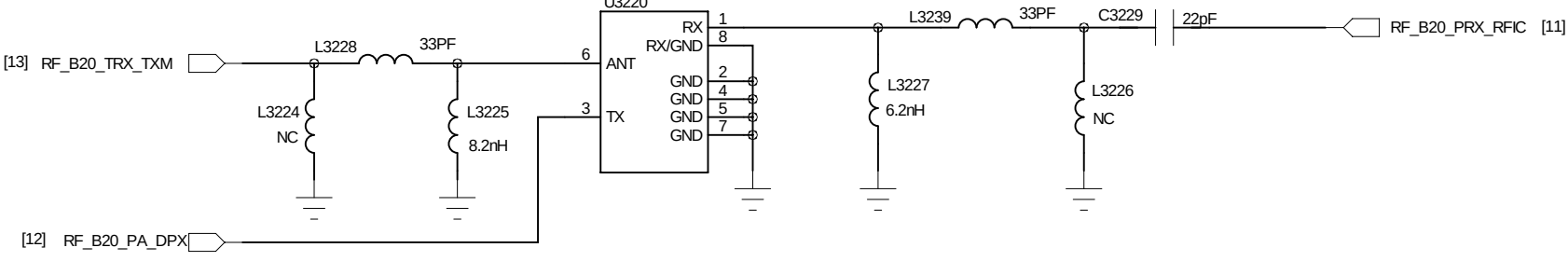
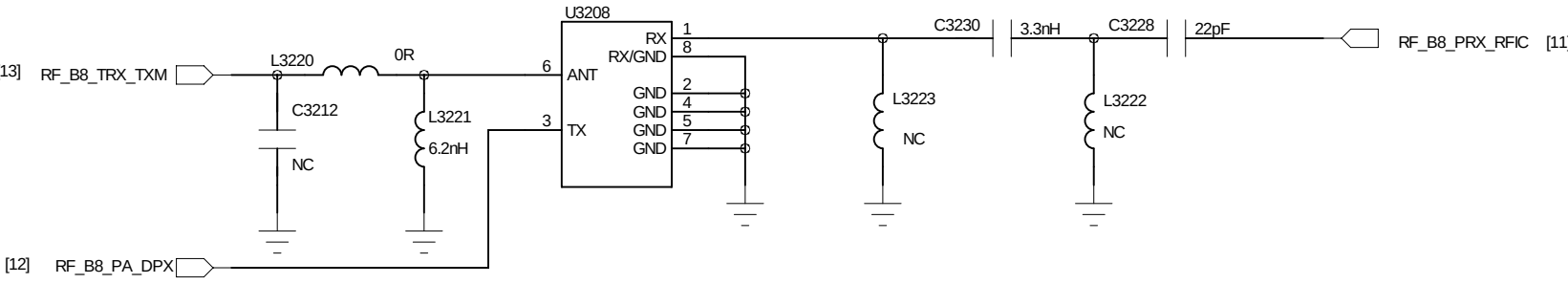
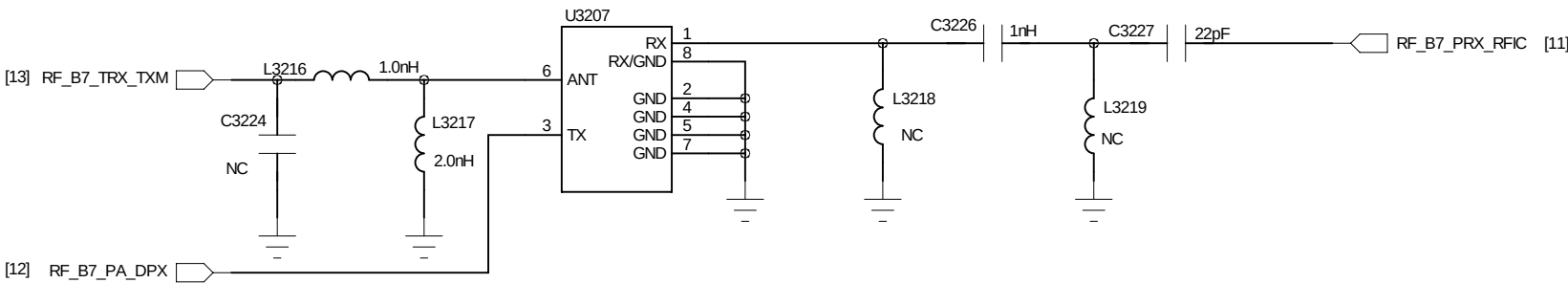
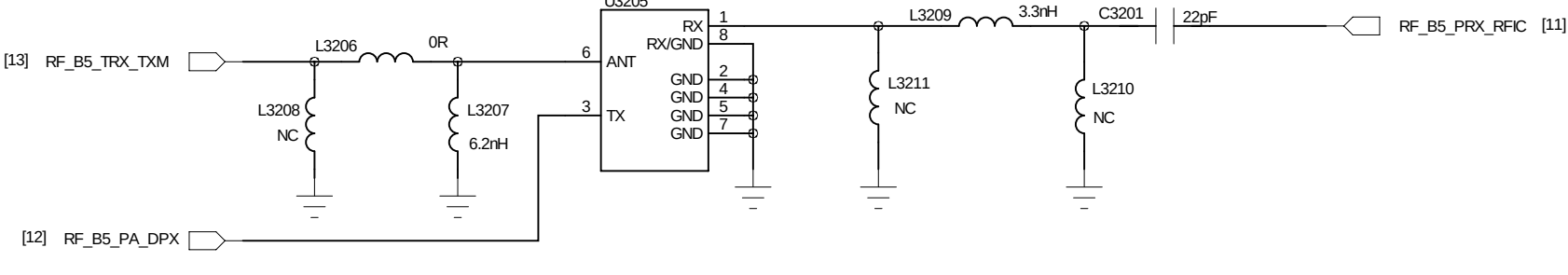
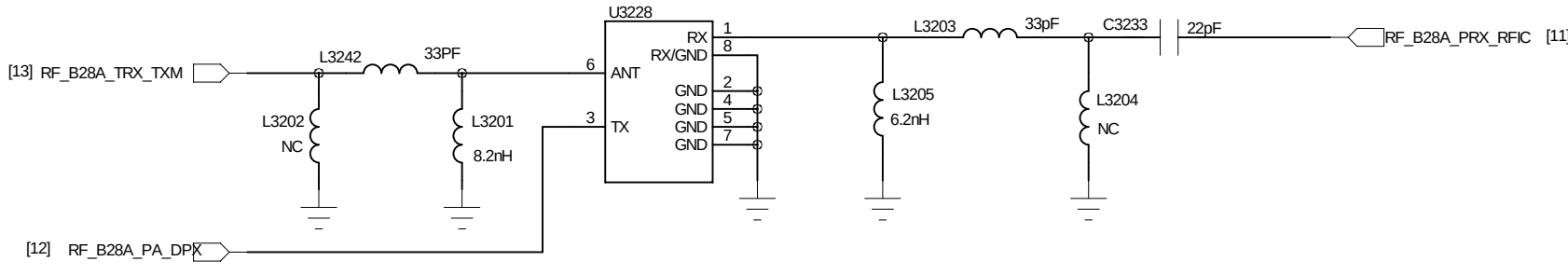
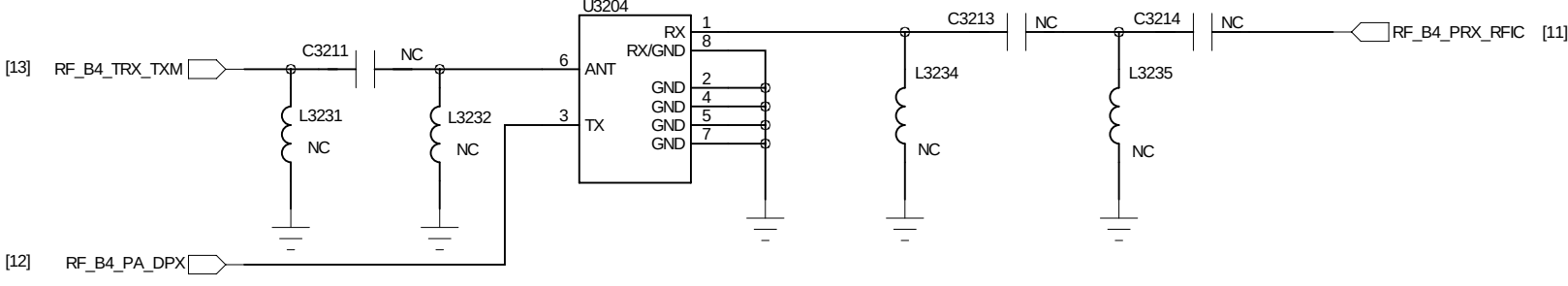
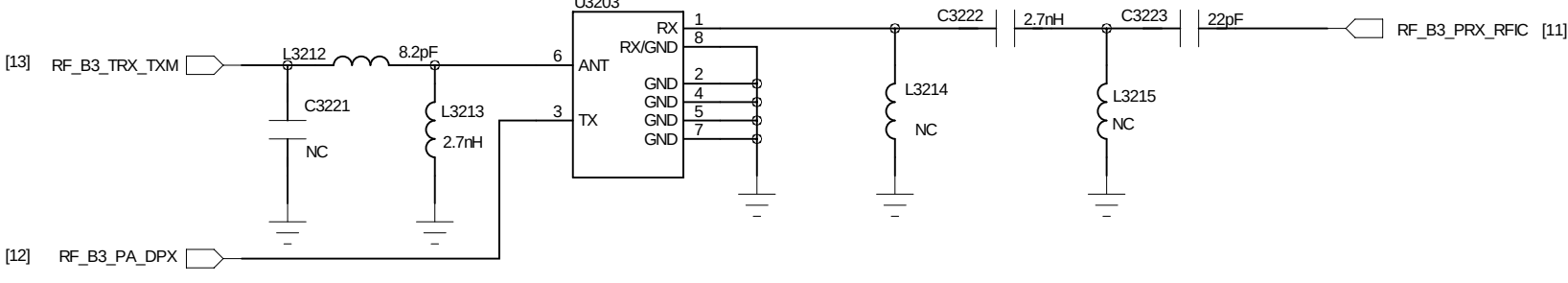
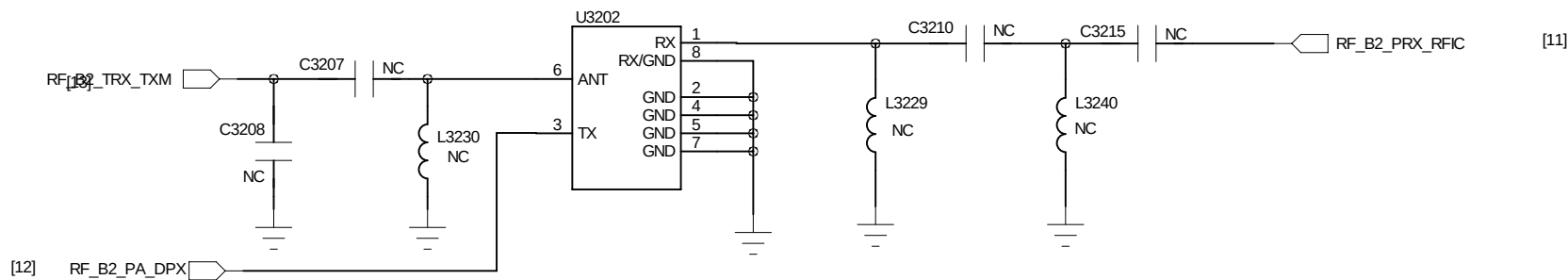
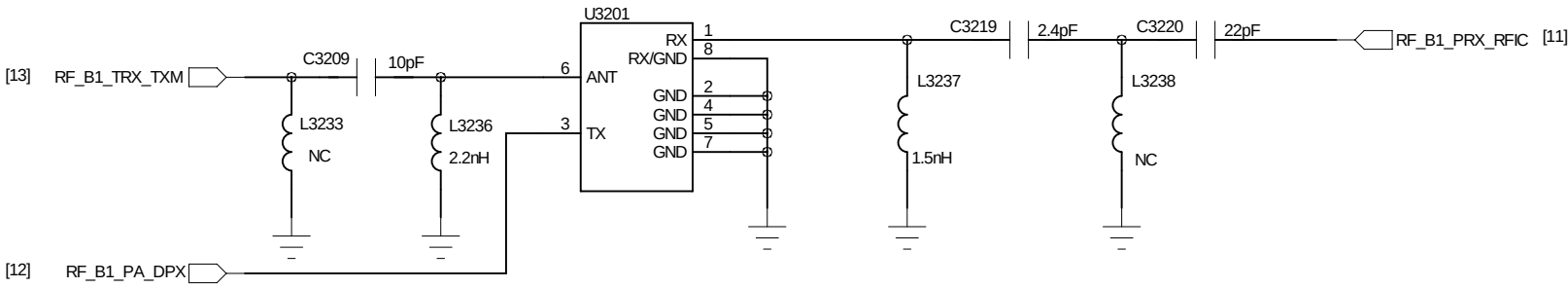
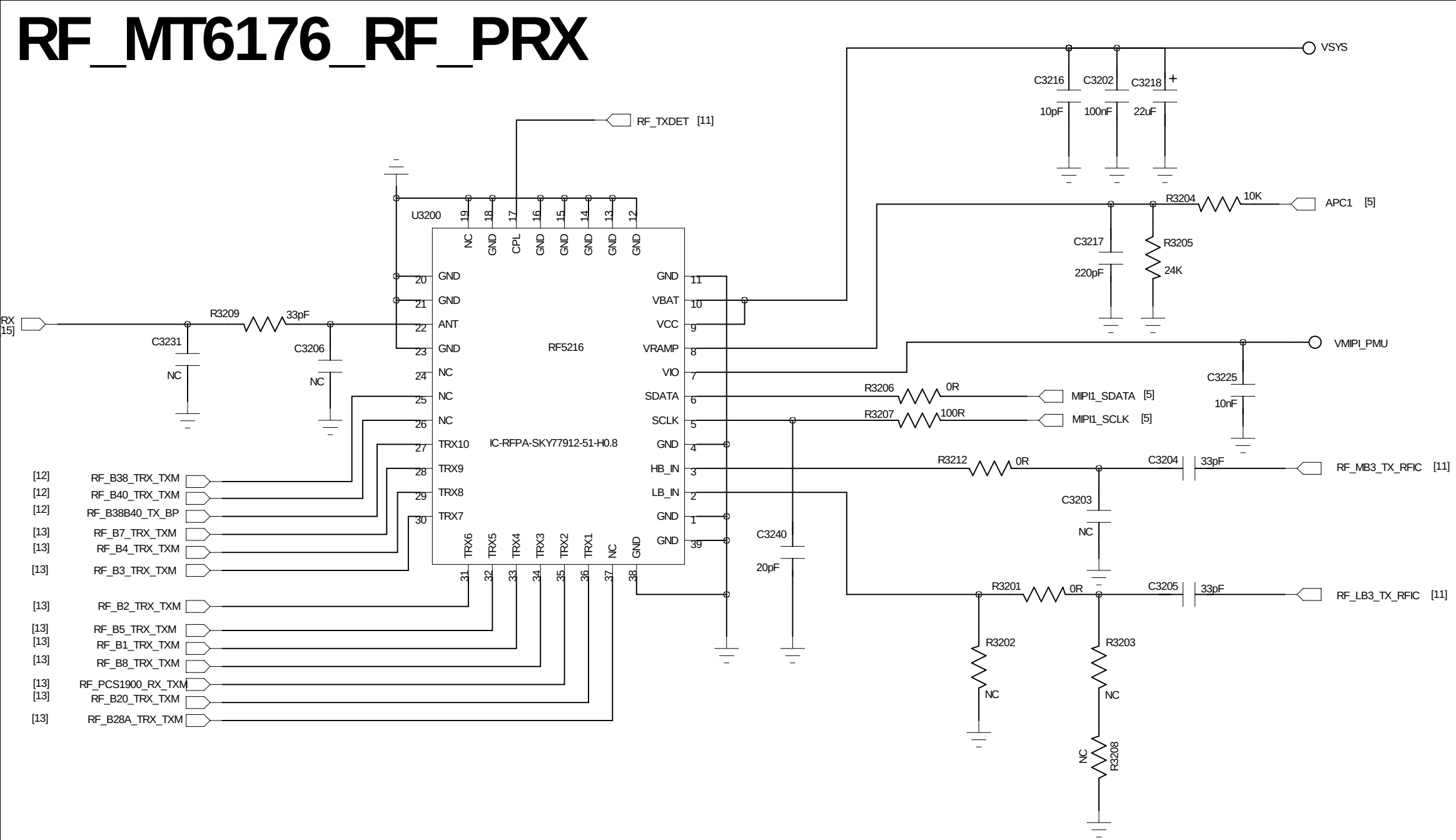
REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:



COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 31_RF_MT6176_RF_TX		VERSION: V1.1	SHEET: 12 OF 23
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RF_MT6176_RF_PRX

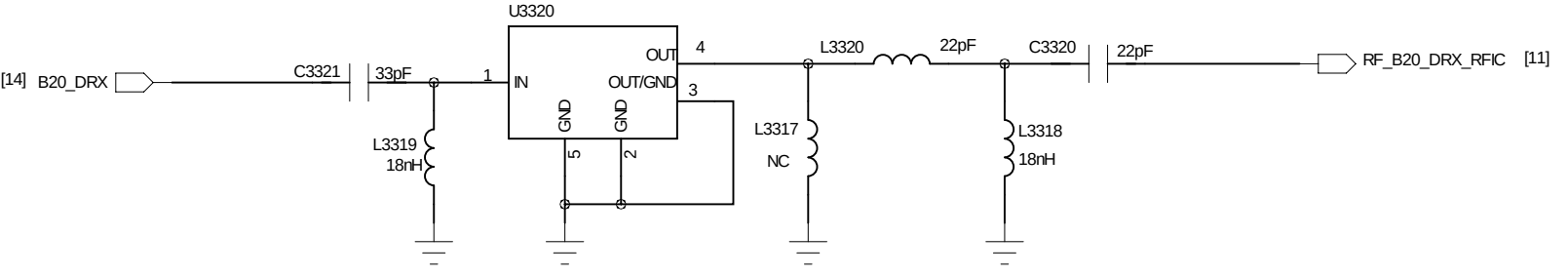
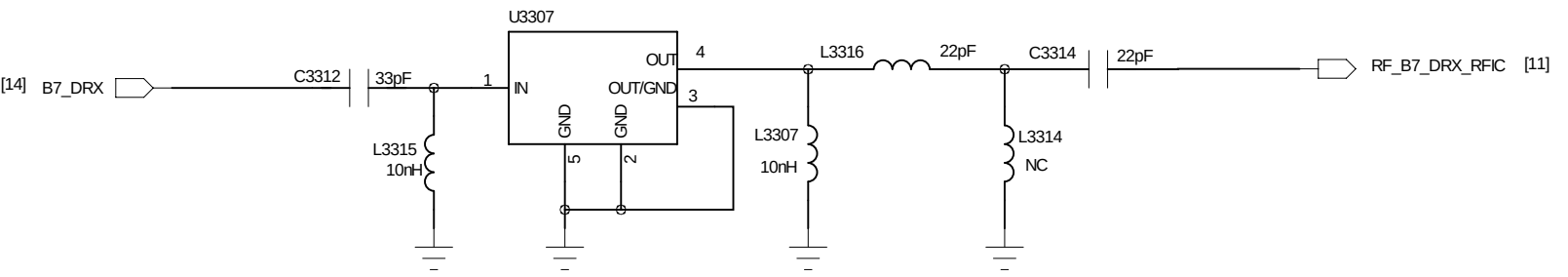
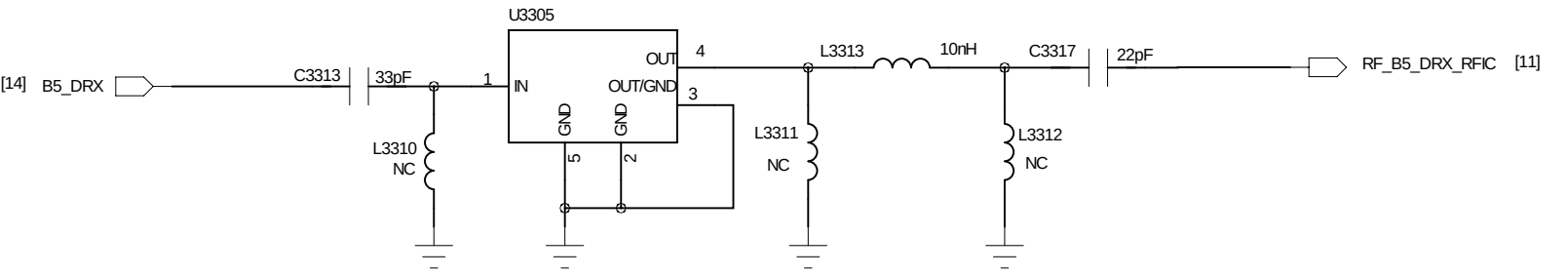
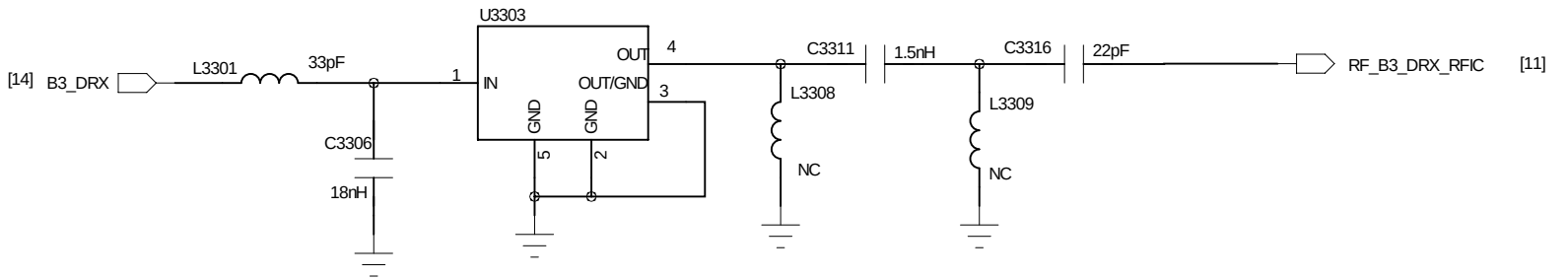
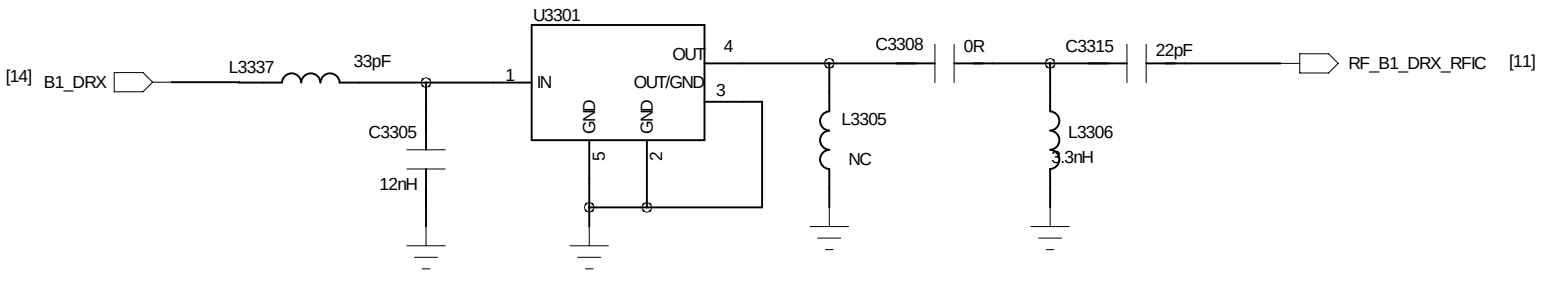
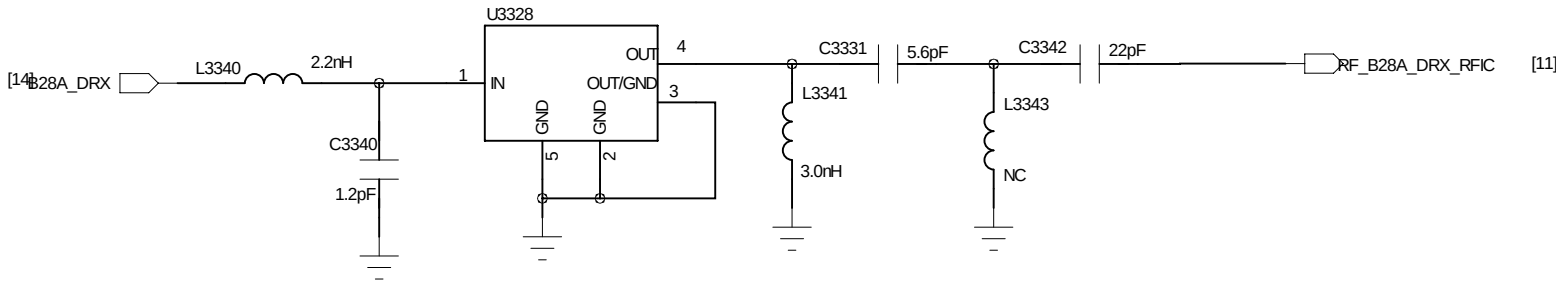
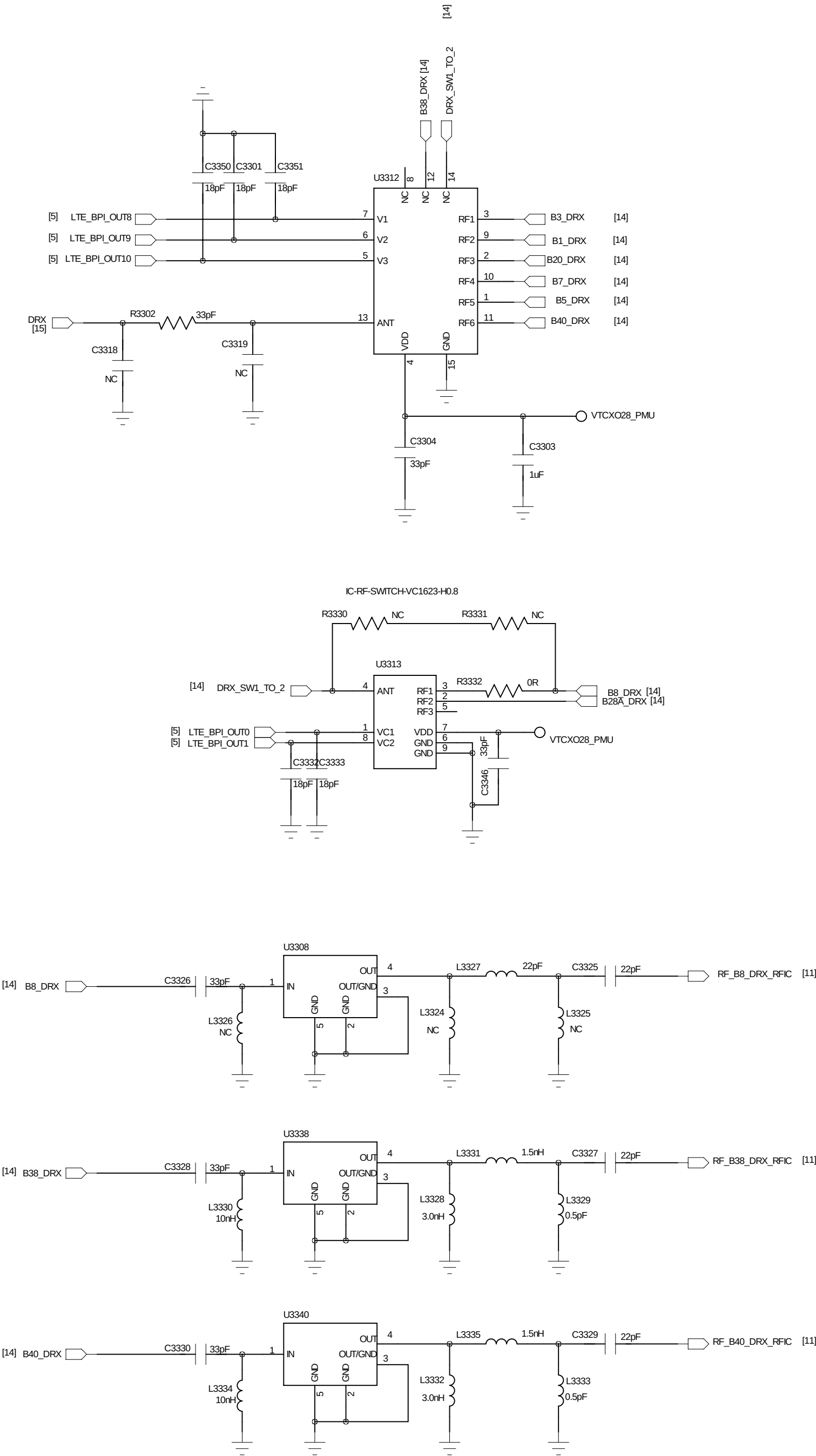
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LTR	ECO NO.	APPROVED:	DATE:



COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 32_RF_MT6176_RF_PRX		VERSION: V1.1	SHEET: 13OF 23
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RF_MT6176_RF_DRX

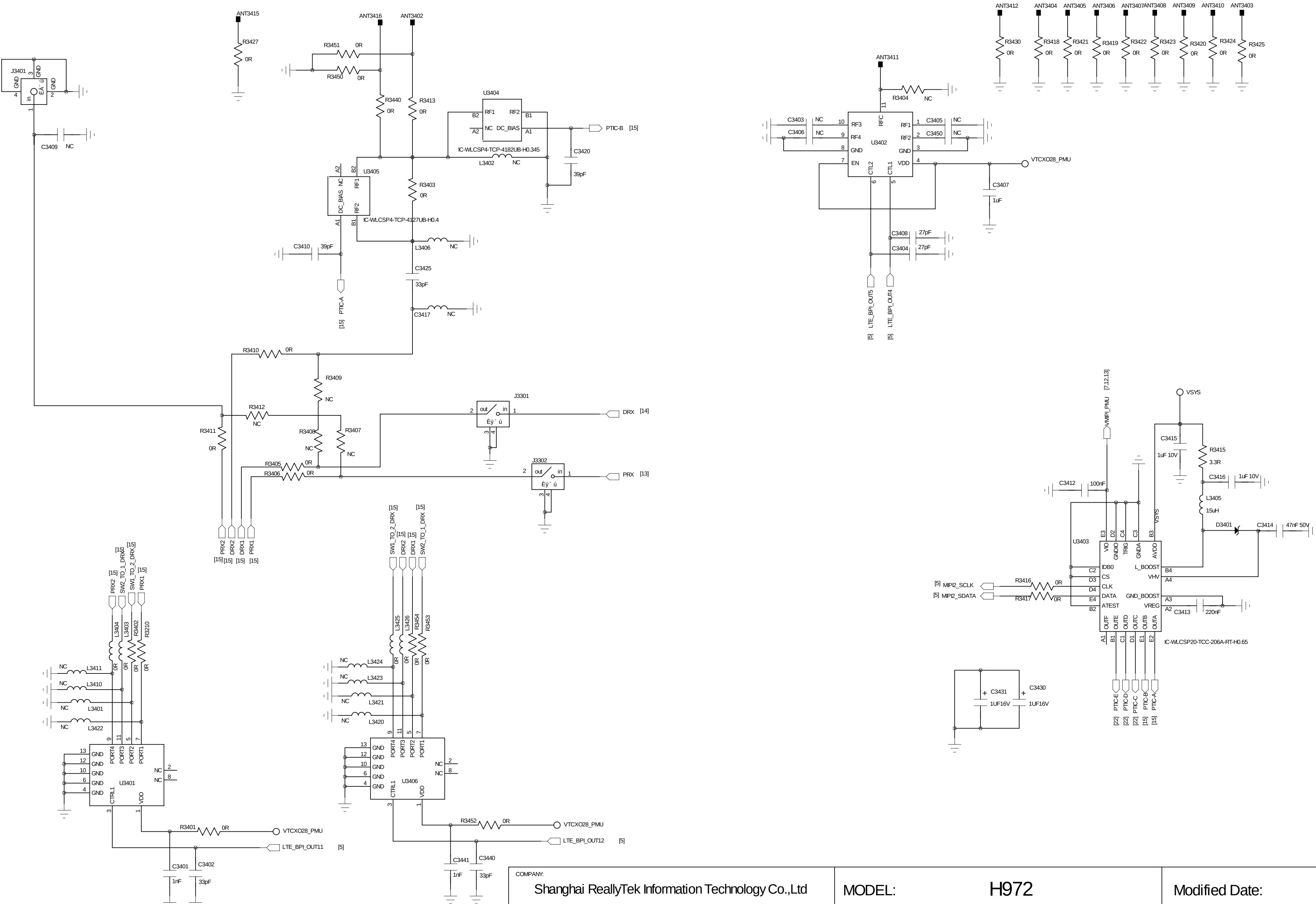
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LTR	ECO NO.	APPROVED:	DATE:



COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 33_RF_MT6176_RF_DRX		VERSION: V1.1	SHEET: 14 OF 23
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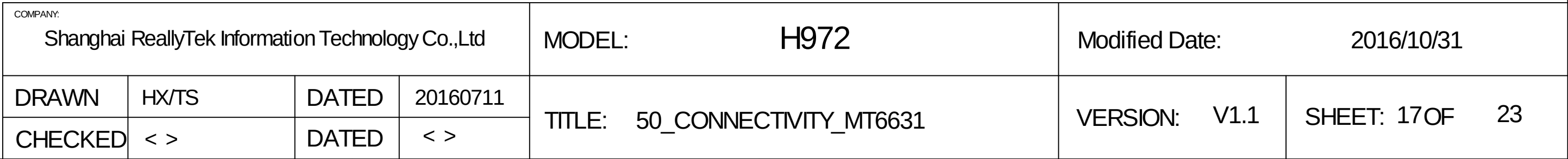
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LTR	ECO NO.	APPROVED:	DATE:



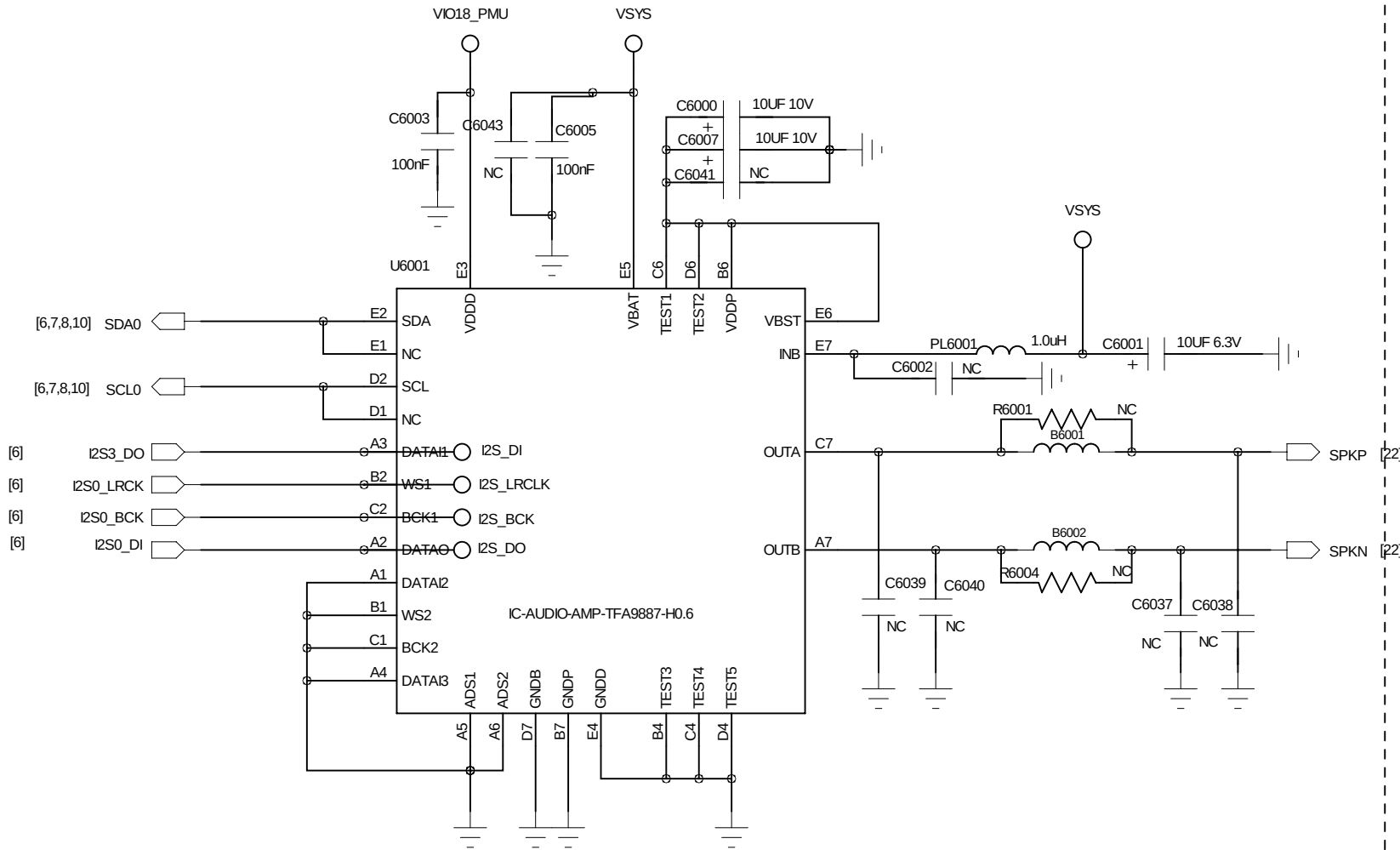
COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
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LTR	ECO NO.	APPROVED:	DATE:

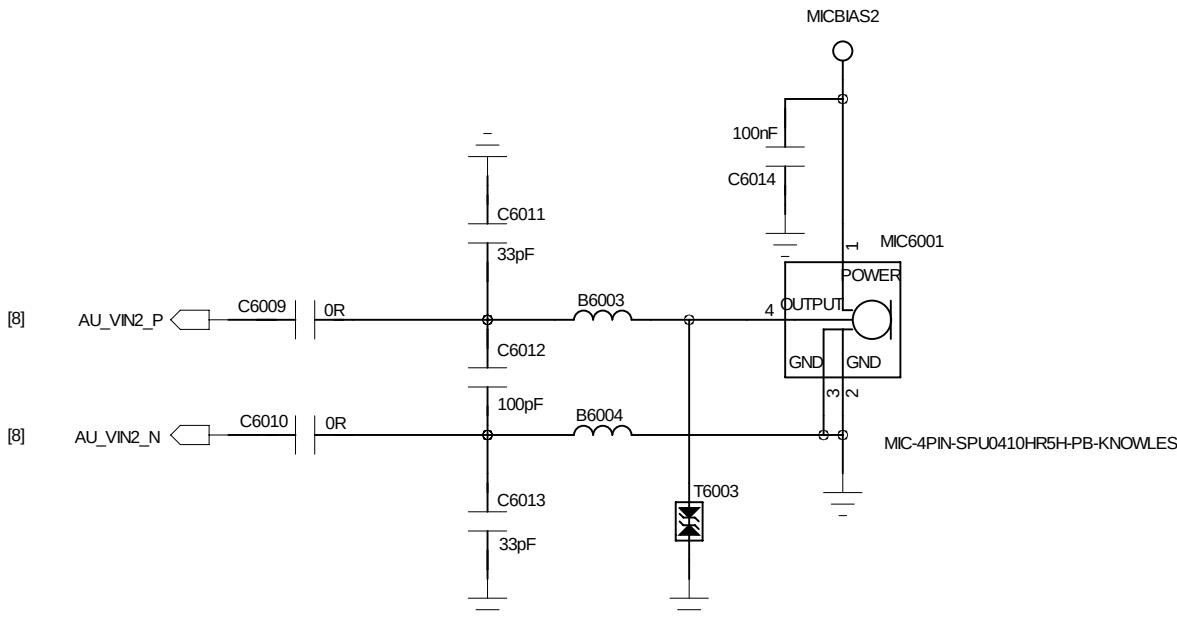


PERI_AUDIO_IO

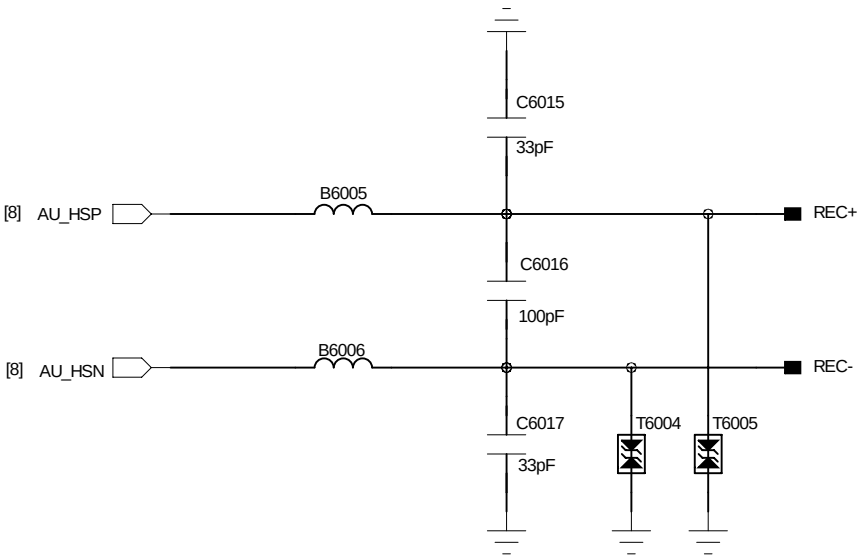
Smart PA



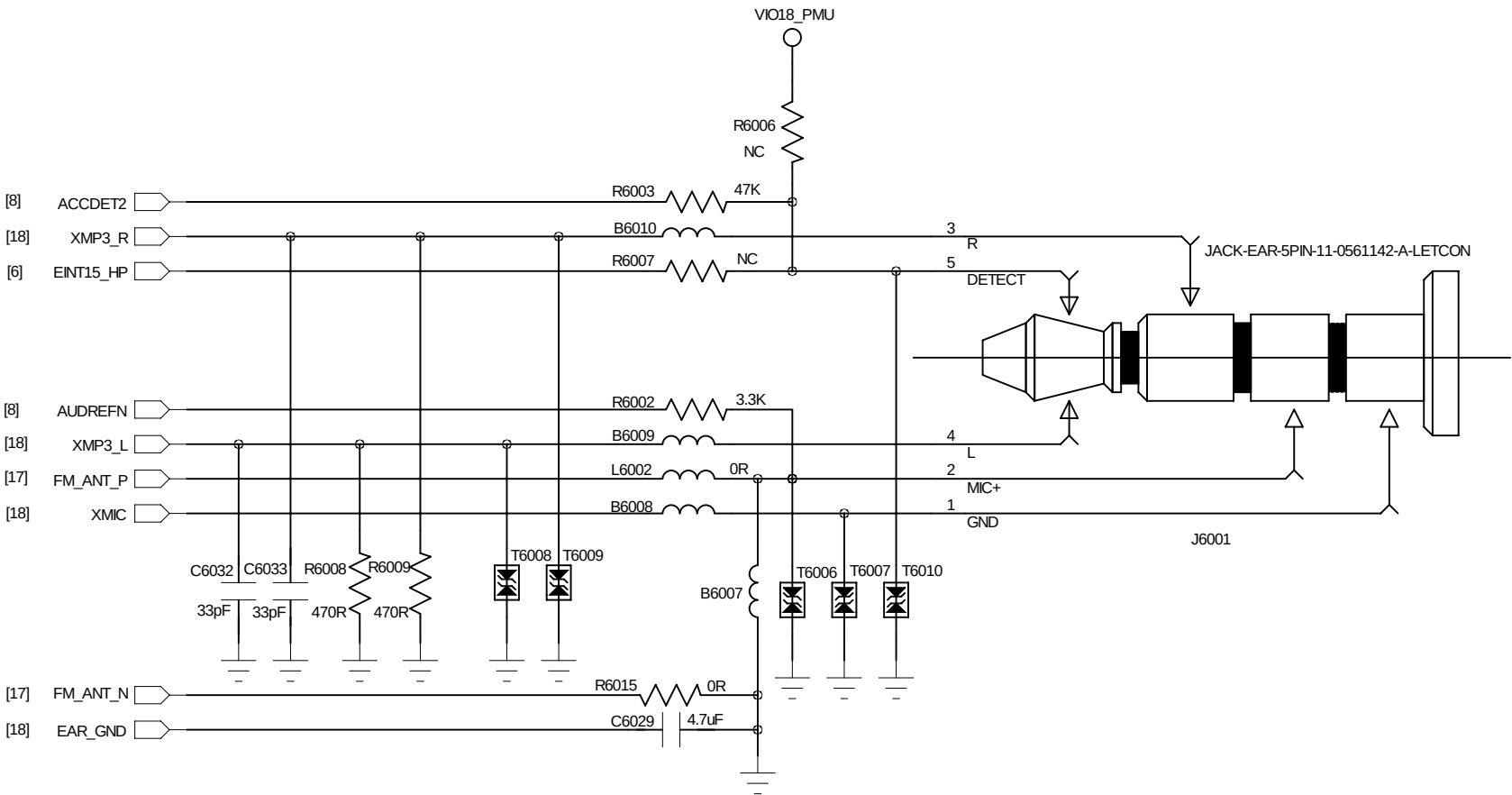
De-noise MICOPHONE1



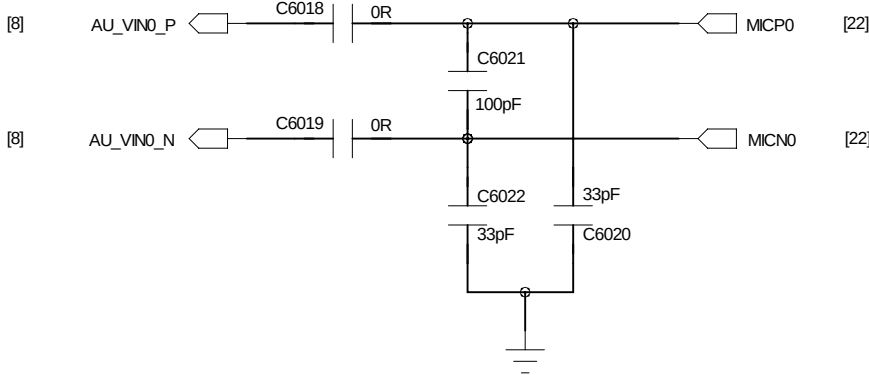
Receiver



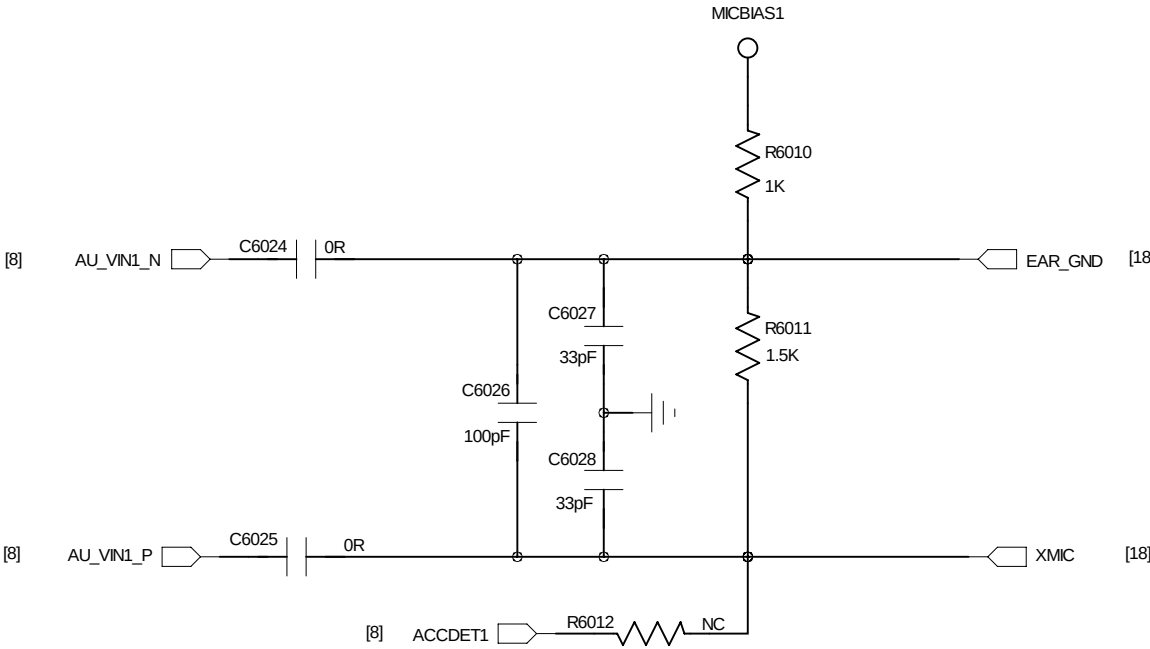
3.5mm Audio Jack



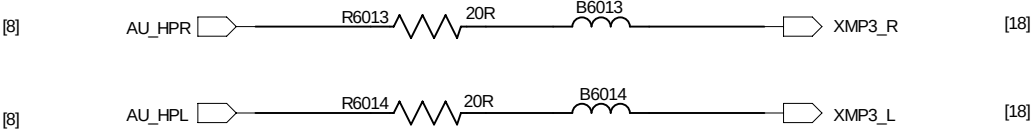
Main MICOPHONE



Earphone MICPHONE



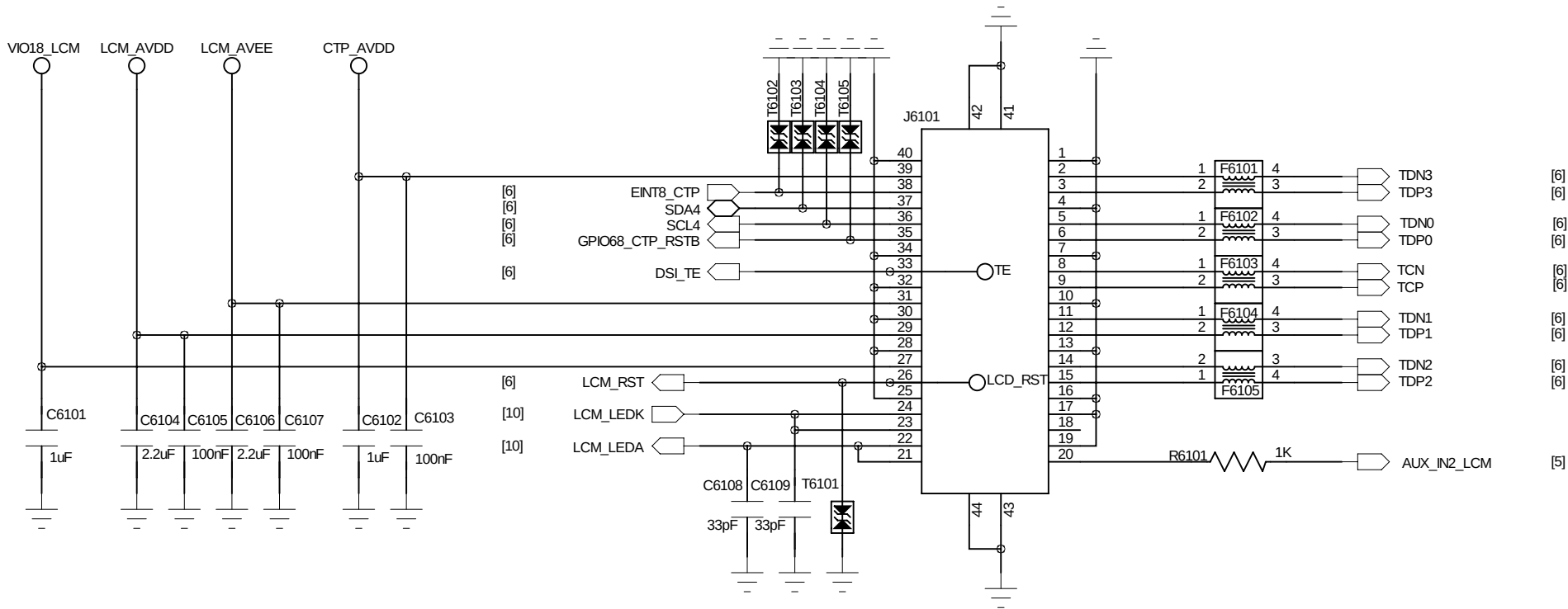
Earphone Receiver



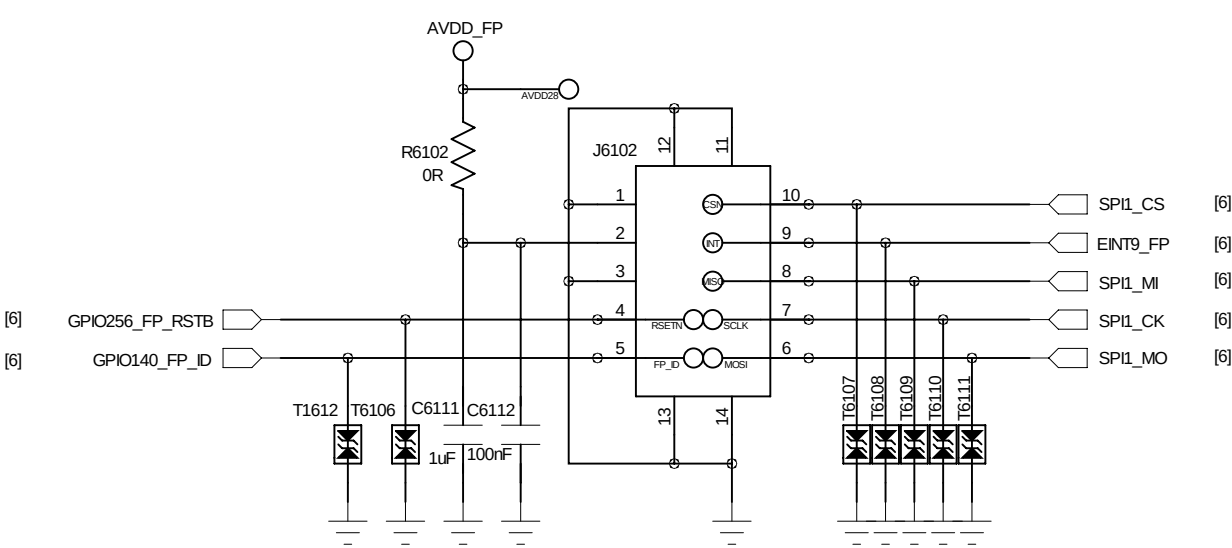
COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
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REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

LCD_CTP_INTERFACE



FINGER PRINT

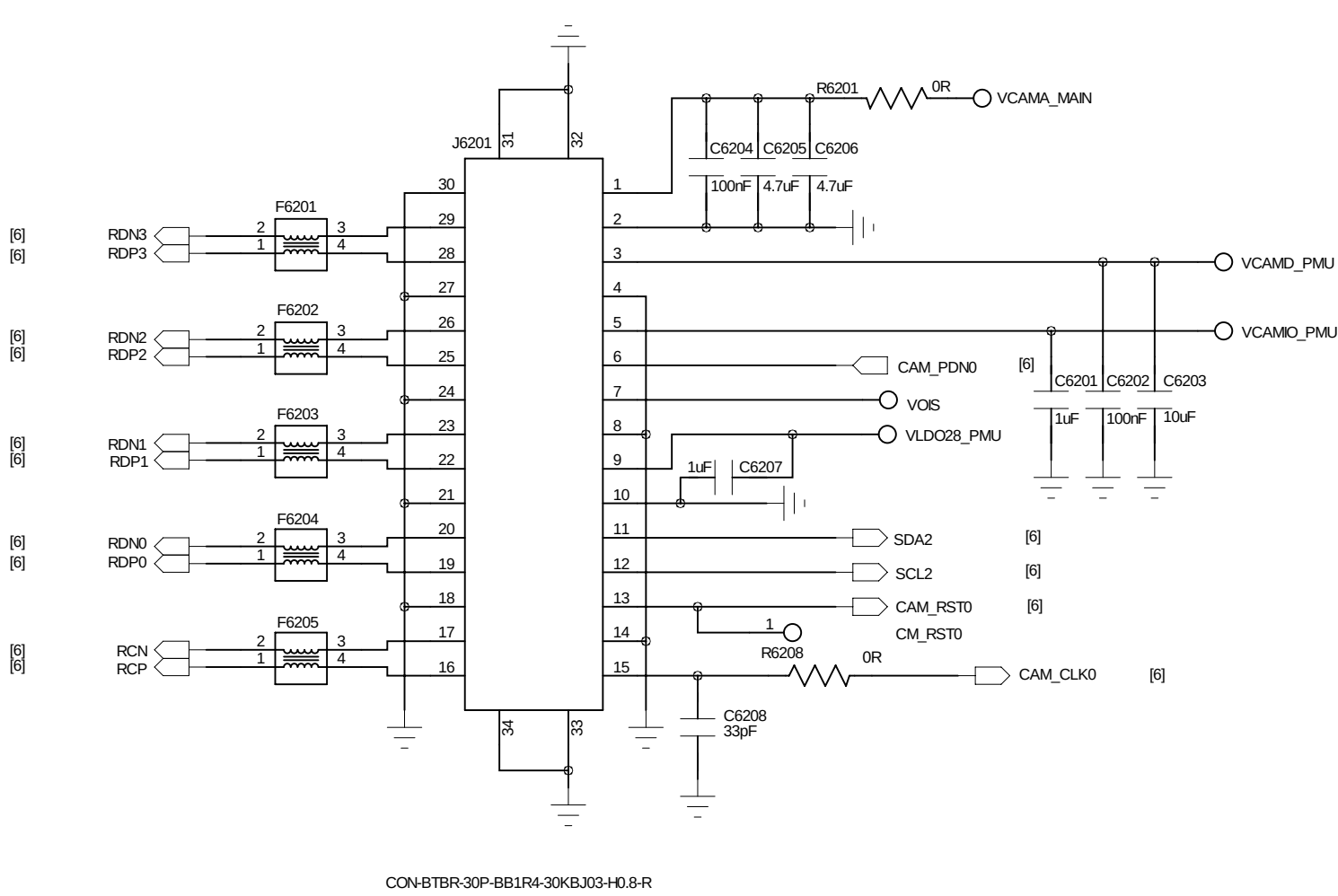


COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
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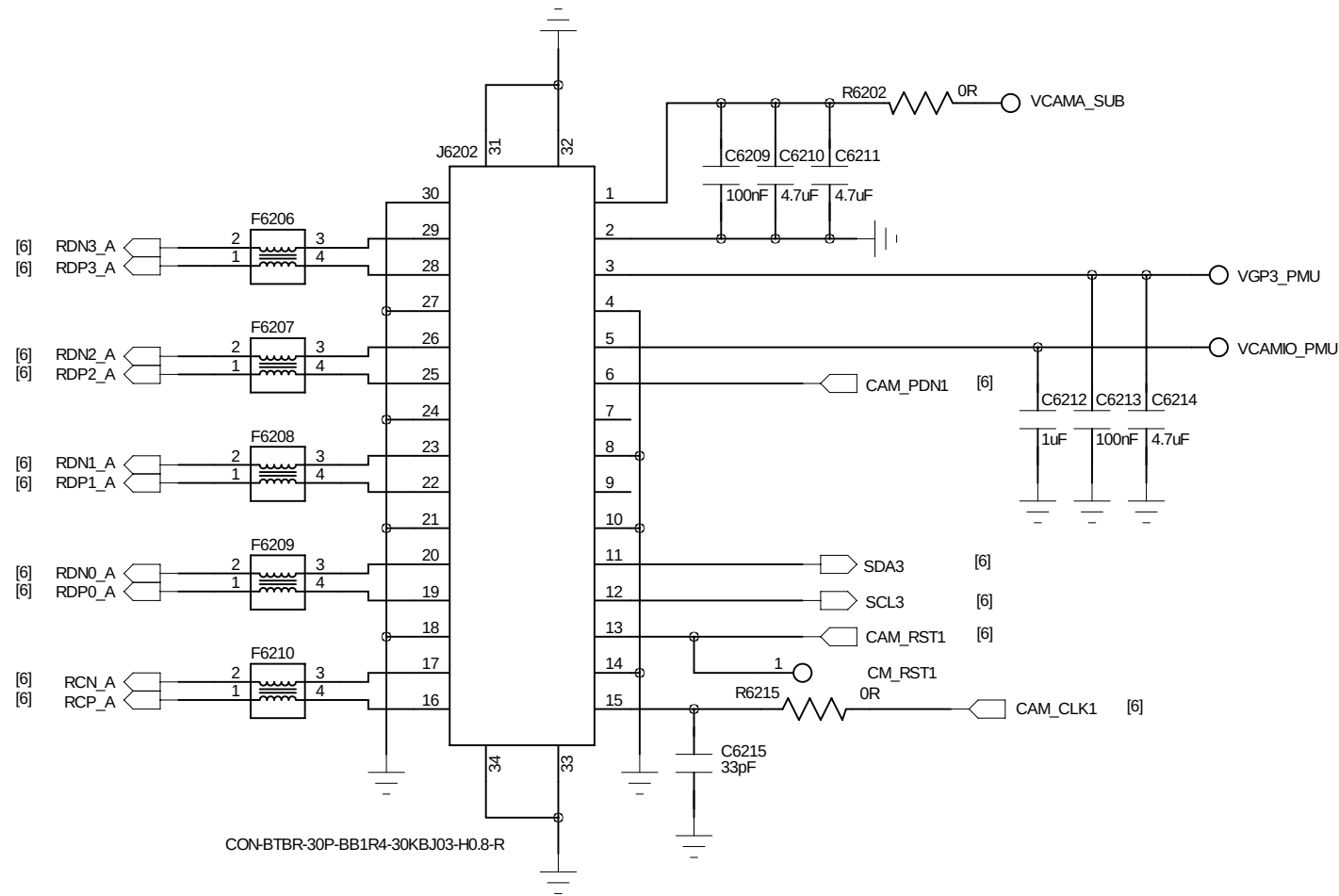
PERI_CAMERA

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

Rear Camera (21M+AF)



Front Camera (13M+FF)

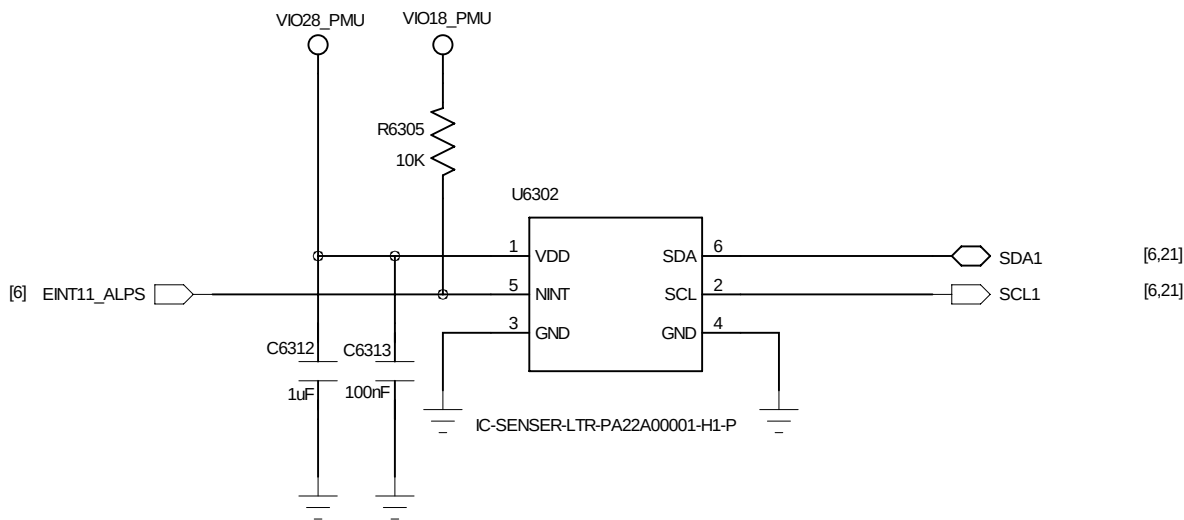


COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 62_PERI_CAMERA		VERSION: V1.1	SHEET: 20 OF 23
CHECKED	< >	DATED	< >				

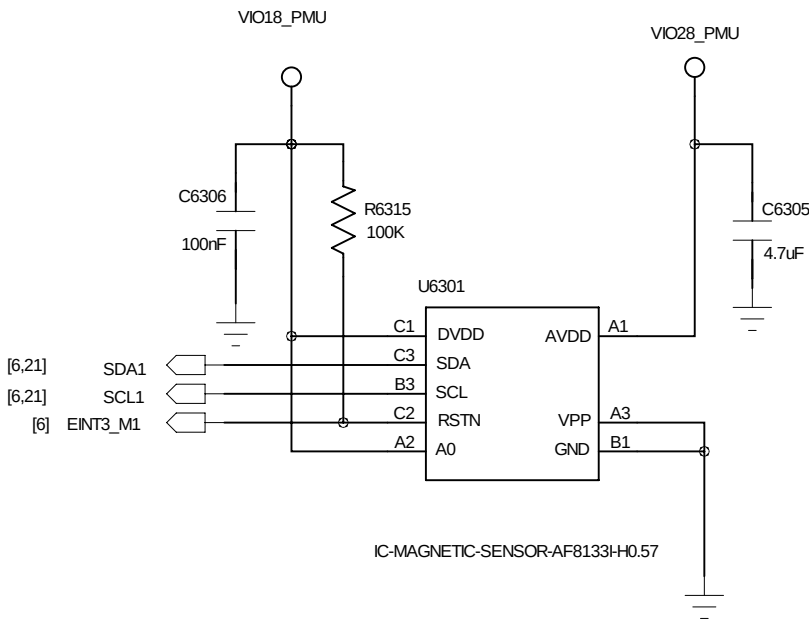
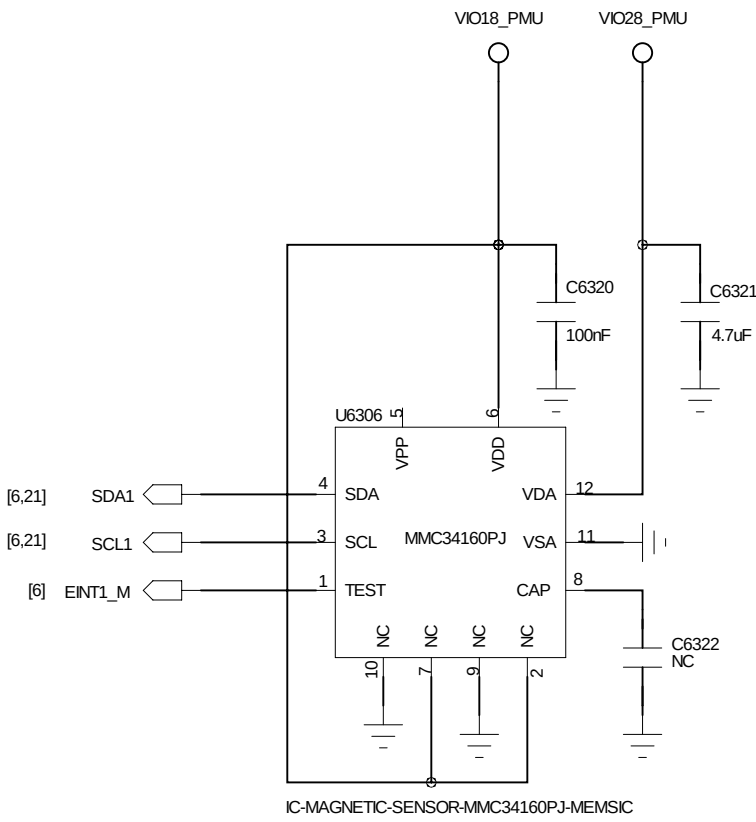
PERI_SENSORS

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

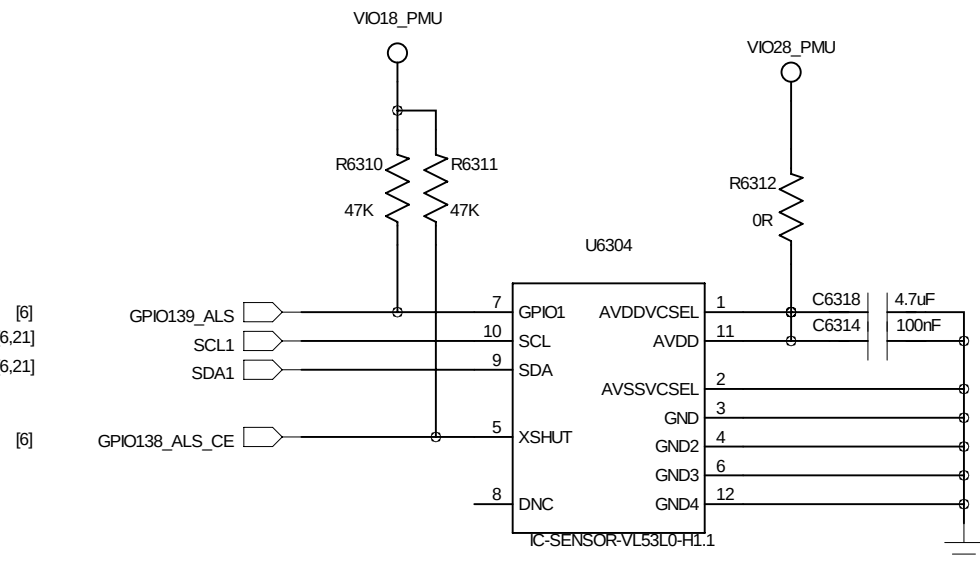
AL& PS Sensor



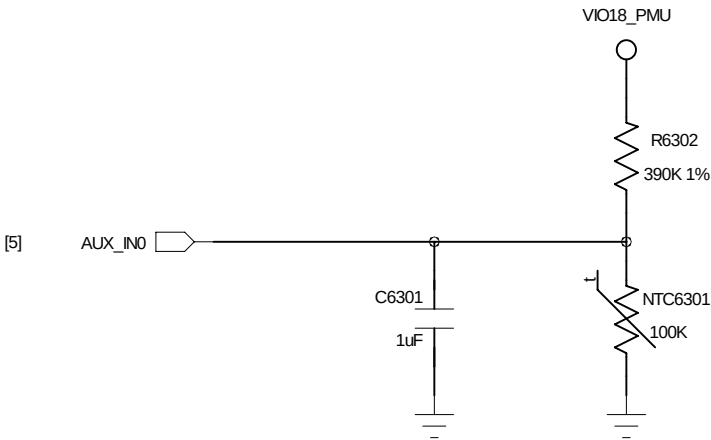
M-Sensor (COMPASS)



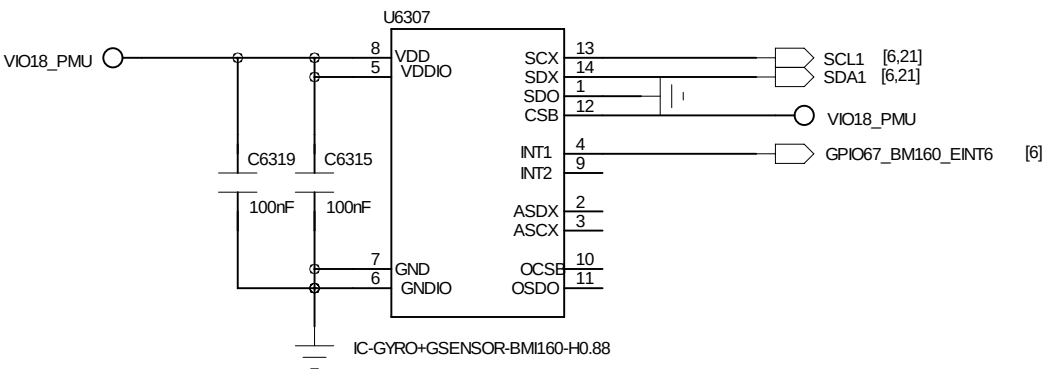
ALS Sensor



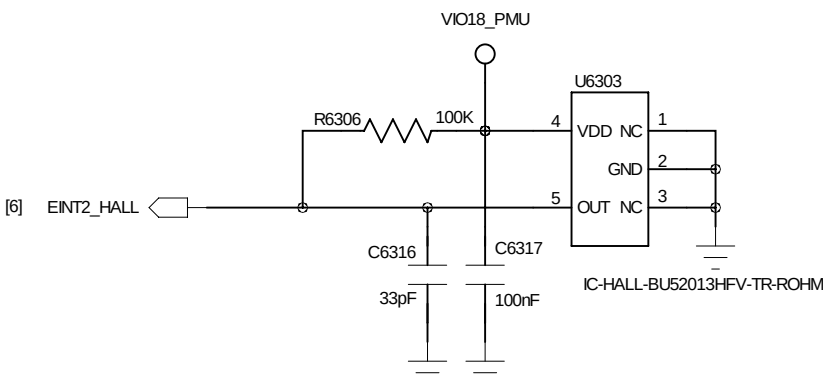
Thermister



Gyro Sensor



Unipolar HALL

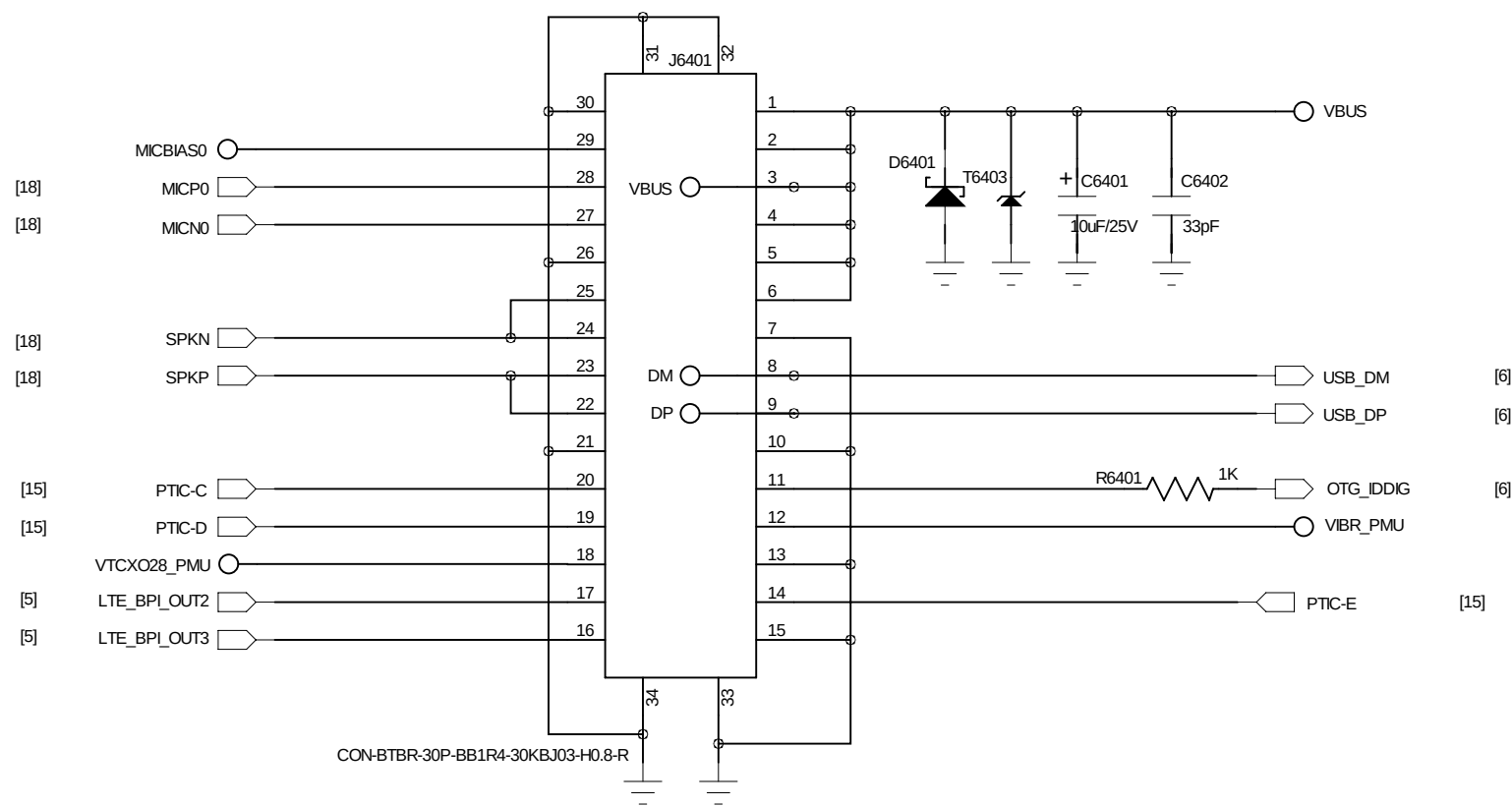


COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 63_PERI_SENSORS		VERSION: V1.1	SHEET: 21 OF 23
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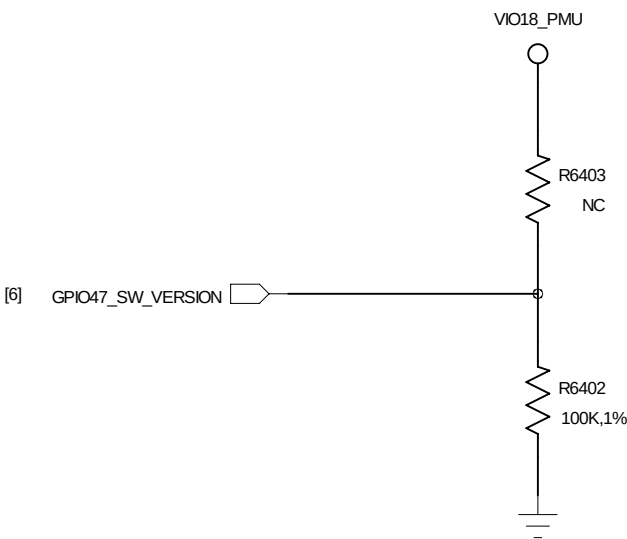
PERI_EXCH_IO

REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

EXCHANGE IO



SW_VERSION



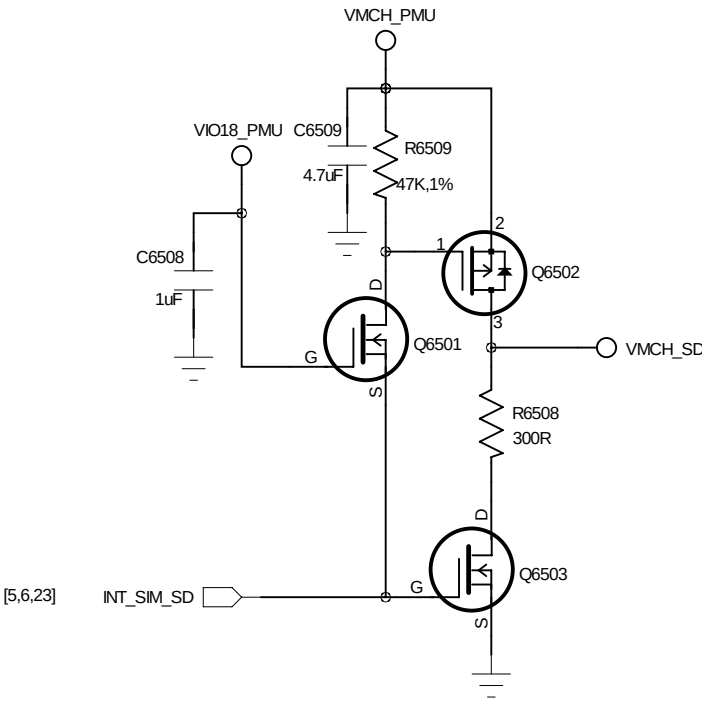
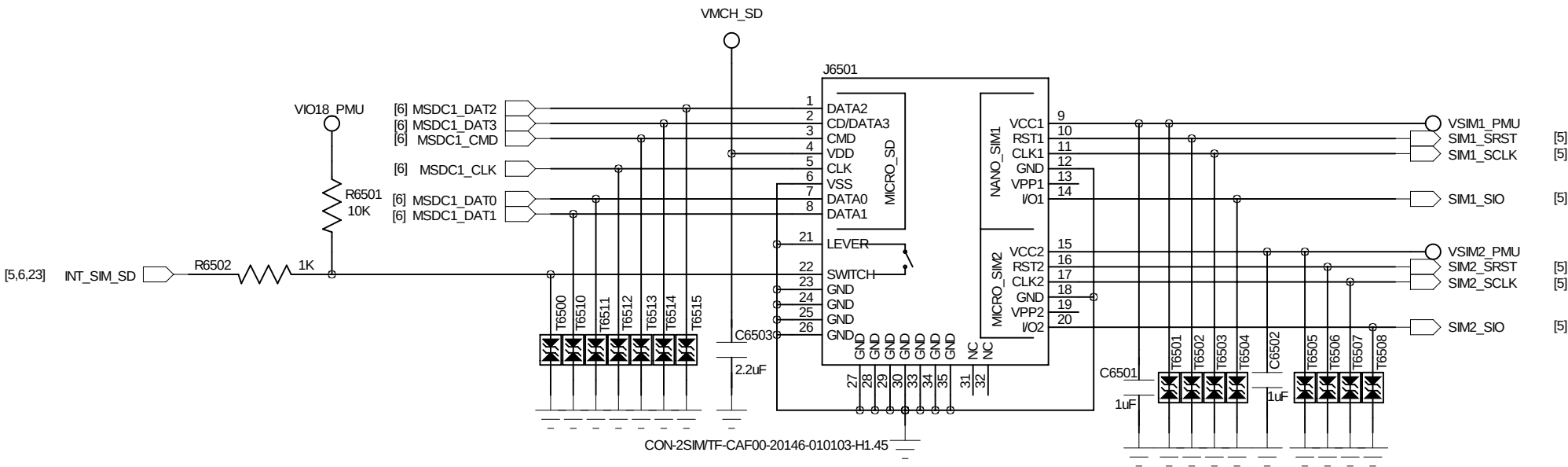
COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 64_PERI_EXCH_IO		VERSION: V1.1	SHEET: 22 OF 23
CHECKED	< >	DATED	< >				

PERI_SIM_SD_KEYPAD

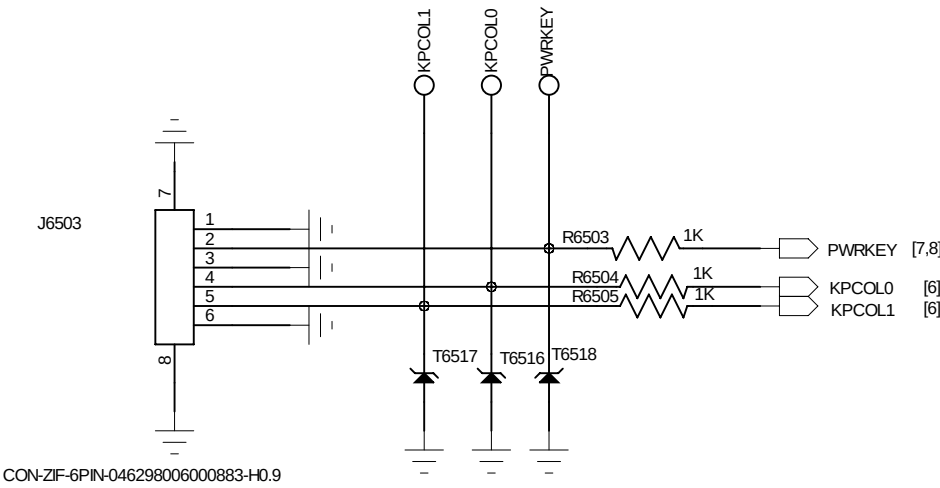
REVISION RECORD			
LTR	ECO NO.	APPROVED:	DATE:

SD POWER CONTROL

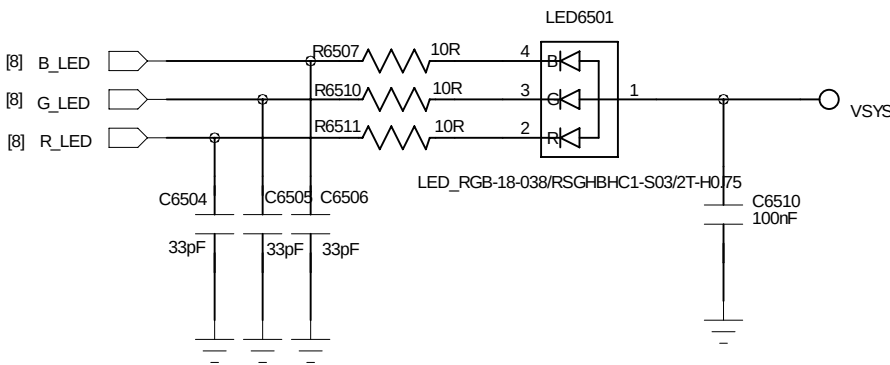
SIM_SD CARD



Power Key&Volume Key



LED



COMPANY: Shanghai ReallyTek Information Technology Co.,Ltd				MODEL: H972		Modified Date: 2016/10/31	
DRAWN	HX/TS	DATED	20160711	TITLE: 65_PERI_SIM_SD_KEYPAD		VERSION: V1.1	SHEET: 23OF 23
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